Timing Search[†]

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Abstract

Why do different industries fluctuate at different frequencies? I develop a model with *timing search*: agents choose when to enter a market under frictions that penalize rapid adjustment. The indifference condition across entry dates prevents the industry from settling at its optimum—entry must persist past optimal scale, forcing a return—and cycles arise endogenously at frequency $\omega = \sqrt{\varphi/\mu}$, pinned by profit-landscape curvature (φ) and matching-congestion friction (μ). Cycle frequency and the welfare cost of cycles have disjoint primitives: market structure (φ, μ) sets frequency, while shock variance and Rayleigh bottleneck dissipation set welfare. Empirically, geographic concentration significantly predicts cycle length in the predicted direction across 16 manufacturing industries and 10 NAICS supersectors. Suggestive additional diagnostics include BDS oscillation features, post-shock periodicity at industry-specific frequencies, and a post-COVID overshoot.

Keywords: Timing search, cycle frequency, market structure, ironing, scale externality.

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1 Introduction

Why do industries fluctuate at different frequencies? Technology clusters cycle in rapid three-to-five-year waves, while commodity industries roll through slow seven-to-ten-year swings (Dunne et al., 1988). Standard business-cycle frameworks derive cycle properties from a common shock process (Kydland and Prescott, 1982a) or policy rule (Christiano et al., 2005), leaving this cross-industry variation unexplained. This paper introduces *timing search*—agents choosing *when* to enter a market under frictions that prevent instantaneous adjustment—and shows that each industry has a *natural cycle length* determined by two market-structure primitives: the curvature of its profit landscape and the timing friction in its entry-matching process. After a shock displaces the industry from optimal scale, the economy does not return monotonically. Instead, it swings back and forth around optimal scale, with entry waves followed by periods of net contraction, until bottleneck friction in the entry-matching pipeline gradually pulls it to rest. The length of each swing depends on the market-structure primitives; the depth depends on the shock variance, the Rayleigh friction coefficient, and the profit-landscape curvature. In closed form,

$$\omega \;=\; \sqrt{\varphi/\mu}\,, \tag{1}$$

where ω is the cycle frequency (the period is $2\pi/\omega$), φ is the curvature of the profit landscape—how sharply profits fall when the industry is too large or too small—and μ is the timing friction arising from congestion in the entry process: when the entry pipeline carries a high replacement flow, adjustment becomes slower and more costly. Both primitives are cross-sectionally measurable. The frequency ω does not depend on the shock process, on policy rules, or on how big a displacement the industry has suffered. It is a property of how firms in the industry interact with one another, not of the macroeconomic environment.

The cycles do not rely on any specific functional form. They arise whenever profits are hump-shaped in the number of active firms (scale externalities below the peak, congestion above) and rapid adjustment is costly. Under these conditions, the equilibrium indifference requirement rules out a quiet return to optimal scale. If the industry tried to drift back slowly, dates near the optimum would combine high profits with low adjustment cost, so agents would crowd those dates until the advantage disappeared. The economy is therefore unable to come to rest gradually; the only

equilibrium-consistent path is one that keeps moving past the optimum, and the resulting motion is cyclical.

The paper's Gaussian specification is important not for the cycles themselves but for their *exact solvability*. Under Gaussian profits and the exponential-quadratic timing friction, the equilibrium reduces to a conservative harmonic oscillator. The cycle period is exactly $2\pi\sqrt{\mu/\varphi}$ for any size of displacement (Theorem 1). This closed-form equilibrium statement is the paper's theoretical centerpiece. For general hump-shaped profits, the formula is approximate. The cycle length picks up a small amplitude-dependent correction, bounded below 1.6% for empirically relevant displacements (Proposition 4).

Convex congestion in entry processing turns the oscillator into the operative equilibrium. Without dissipation, cycles would persist forever. In practice, the bottleneck through which entry flow is processed (permits, specialist labor, construction supply chains) imposes a damping force proportional to the flow, summarized at the macroscopic level by a Rayleigh friction coefficient $\chi > 0$, distinct from the per-firm entry cost κ. Each swing dissipates energy at rate $\chi\dot{n}^2$, so the industry responds to shocks like a bell that has been struck: it rings at its natural pitch, and the ringing gradually decays back to rest at optimal scale. Ongoing shocks re-excite the oscillations. The stationary standard deviation of $(n - n_p)$ is $\sigma_x = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$, where $\eta = \chi/\mu$ is the effective decay rate (Proposition 7).

Under linear utility, the welfare cost of these fluctuations simplifies to

$$\mathcal{L} = \sigma_\varepsilon^2/(2\chi),$$

independent of both market-structure primitives φ and μ, and independent of the per-firm cost κ. In the model, industrial policy acts on frequency through (φ, μ) and on welfare through $(\sigma_\varepsilon, \chi)$, with the two channels orthogonal (Section 6.2). Whether a specific real-world policy reduces κ (the per-firm cost), χ (the Rayleigh friction coefficient), or both is an institutional question discussed in Section 6.3.

Cycle frequency and the welfare cost of cycles therefore decouple in equilibrium. How fast an industry swings back and forth is a pure market-structure property, while the welfare cost of those swings depends on shock variance and the friction that damps the motion. Frequency and welfare cost are governed by disjoint sets of structural primitives, $\{\varphi, \mu\}$ and $\{\sigma_\varepsilon, \chi\}$ respectively. Cycle severity is the intermediate

observable: with $\eta = \chi/\mu$, the matching friction μ cancels in the stationary standard deviation $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, so severity depends on $(\sigma_\varepsilon, \chi, \varphi)$ but not on μ. This frequency-welfare separation is the paper's core theoretical result.

The separation sets timing search apart from standard frameworks. In a neoclassical AR(1) economy, a bigger shock takes longer to recover from, so magnitude and duration are tied together through the persistence parameter, and industries hit by common shocks should share a common cycle length. U.S. industries do not satisfy this prediction: spectral peaks range from 2.6 to 6.6 years in industrial production and from 2.4 to nearly 10 years in establishment entry. Timing search accommodates this heterogeneity because the ironing condition—the temporal analog of the equal-profit condition in Burdett and Mortensen (1998)—ties cycle length to market structure rather than to the shock process.

The separation of cycle length from cycle severity has a sharp mechanical interpretation. The equilibrium dynamics of timing search take the form $\ddot{x} = -(\varphi/\mu)\, x$ for $x = n - n_p$, which is the equation of a harmonic oscillator with natural frequency $\omega = \sqrt{\varphi/\mu}$. This frequency is a property of the equilibrium *linearization*—an eigenvalue of the dynamics at rest—rather than a feature of any particular trajectory. An industry that sits exactly at optimal scale and is never hit by shocks would remain at rest indefinitely, yet the frequency ω would still be embedded in the structure of the equilibrium even though the amplitude of visible motion would be zero. A shock, or any initial deviation from n_p, excites oscillation at exactly this structural frequency. In this sense every industry has a natural pitch, and whether the industry is currently ringing—and how loudly—is a separate question determined by shocks and dissipation. Market structure determines the pitch; the shock environment determines whether the bell has been struck.

Why does the industry overshoot rather than settle? The same indifference requirement that fixes the cycle frequency also pins a non-trivial rate of entry along the approach to optimal scale, which becomes momentum that cannot be discarded at n_p without reopening a profitable date. The industry must pass through. Section 3 formalizes the argument; Theorem 1 states the closed-form period.

The core contribution is the frequency-welfare separation. Cycle frequency depends only on the market-structure primitives (φ, μ), while the welfare cost of cycles depends only on the shock variance σ_ε^2 and the Rayleigh friction coefficient χ; cycle severity is the intermediate observable, depending on $(\sigma_\varepsilon, \chi, \varphi)$. This is not a modeling

assumption but an equilibrium implication of a single primitive, the ironing condition. It gives economists new mental categories for thinking about business cycles: when two industries cycle at different frequencies, the explanation can be market structure rather than different shocks or policy exposure; when an industrial policy proposal is evaluated within the model, frequency, amplitude, and welfare respond to overlapping but identifiable parameter channels, conditional on the institutional mapping between the policy and the Rayleigh friction coefficient χ discussed in Section 6.3.

The sharpest instrument of the separation is the closed form $\omega = \sqrt{\varphi/\mu}$, derived as an equilibrium outcome of the ironing condition (Sections 2–3). The result is exact under Gaussian profits and accurate to 1.6% at empirically relevant amplitudes for general hump-shaped profits (Proposition 4; Online Appendix S). The timing friction μ is micro-founded from congestion in entry matching, connecting to the DMP literature (Section 2.3; Pissarides 2000; Petrongolo and Pissarides 2001). A side-by-side simulation comparison with RBC, New Keynesian, and Beaudry et al. (2020) dynamics (Online Appendix O) finds that, in the parameterizations considered, the timing search model is the one that simultaneously produces cross-industry frequency heterogeneity, post-shock periodicity, and overshoot.

The decomposition's directional predictions are supported by U.S. industry data. The headline test is a cross-industry regression of spectral peak frequency on the Ellison and Glaeser (1997a) geographic concentration index across 16 manufacturing 3-digit industries: $\hat{\beta}_\gamma = 0.20$, HC1 SE 0.07, $R^2 = 0.39$ (Section 5). The same positive sign holds in a parallel cross-section of 10 NAICS supersectors using a different data source for both cycle frequency and geographic concentration ($\hat{\beta} = 0.107$, HC1 SE 0.043, $R^2 = 0.39$), indicating that the relationship is not specific to manufacturing. We treat both regressions as sign tests of the directional prediction: the model's exact-magnitude prediction $\beta = 1/2$ is rejected in both samples, with the gap interpretable as either a high specification-noise share in the proxy mapping $\gamma \mapsto \varphi$ or as evidence that geographic concentration captures additional agglomeration channels beyond profit-landscape curvature. Three additional diagnostics, presented as suggestive evidence rather than as headline tests, are consistent with the mechanism: BDS net entry rates exhibit oscillatory features in 10 of 19 sectors, manufacturing and oil oscillate at different industry-specific frequencies ($T = 5.0$ and 3.3 years respectively) after large historical shocks, and 14 of 19 BDS sectors overshoot above their pre-pandemic trend after COVID-19.

Related literature This paper builds on five strands of literature. The first is endogenous business cycles. Beaudry et al. (2020) generate a stochastic limit cycle from financial frictions via Hopf bifurcation. Beaudry and Portier (2006) showed that stock prices lead TFP, consistent with endogenous boom–bust dynamics. Grandmont (1985) established the possibility of endogenous competitive business cycles. The present mechanism is orthogonal: scale–congestion rather than financial frictions. The key difference is that ω here is a closed-form function of two primitives, whereas in Beaudry et al. (2020) it is determined implicitly by a bifurcation condition. The two mechanisms could coexist.

The second is search theory. Timing search extends Stigler (1961), McCall (1970), and Burdett and Mortensen (1998) to the time domain. In standard search, agents choose among contemporaneous alternatives (wages, prices, jobs); here, they choose among dates. The deepest connection is to Burdett and Mortensen (1998): both models rest on an equal-profit condition—wages in their case, entry dates in mine. Online Appendix A develops the formal one-sided problem.

Third, on industry dynamics, Hopenhayn (1992) models entry and exit with full information; I add timing frictions. Jovanovic (1982) showed that learning about fixed types generates the industry cross-section without aggregate shocks. Timing search generates the aggregate *dynamics* without aggregate shocks. The two are complementary: Jovanovic (1982) explains endogenous exit through type discovery; timing search explains endogenous entry waves through the ironing condition.

Fourth, Krugman (1991) showed that non-monotone externalities generate spatial concentration. Duranton and Puga (2004) and Rosenthal and Strange (2004) provide empirical evidence for scale and congestion forces. Ellison and Glaeser (1997a) and Ellison et al. (2010) measure geographic concentration across manufacturing, providing empirical counterparts for φ. I transpose the scale–congestion logic from space to time: combined with timing frictions, it generates temporal patterns (cycles) rather than spatial patterns (clusters).

Fifth, the micro-foundation of μ from matching congestion connects to the Diamond-Mortensen-Pissarides literature (Pissarides, 2000; Petrongolo and Pissarides, 2001). In DMP models, matching frictions generate equilibrium unemployment. Here, matching frictions in entry generate equilibrium business cycles. The formula $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ maps the matching elasticity α to cycle frequency.

2 Environment

The environment has three elements chosen to make the cycle-frequency result precise: a hump-shaped profit landscape that gives the industry a well-defined optimal scale, a matching function for entry that endogenously generates a quadratic adjustment cost, and linear-utility households that let industries evolve independently. Section 2.1 sets up the agents, Section 2.2 the profit landscape, Section 2.3 the timing friction, and Section 2.4 combines them into the reduced-form dynamics that drive the equilibrium.

2.1 Agents

The economy consists of two types of agents: a representative household that consumes and supplies labor, and a fixed mass of potential entrepreneurs who alternate between operating firms and searching for entry opportunities. The household has linear utility over consumption, $U = \int_0^\infty e^{-\rho t} C(t)\,\mathrm{d}t$, and supplies labor perfectly elastically at a fixed wage $\bar{w}$. Linear utility is not merely a simplification: it eliminates wealth effects and cross-industry price feedback, making the single-industry analysis a general equilibrium result (see below). Three ingredients define the entry side of the environment.

Assumption 1 (Entry economy)**.**
A mass $M > 0$ of potential entrepreneurs, each at any instant either active *(operating a firm) or* searching *(a potential entrant). Active firms produce using labor hired from the representative household at wage $\bar{w}$, with technology $y_i = z l_i^\gamma$, $\gamma \in (0, 1)$. Firms exit at Poisson rate $\delta > 0$, returning to the search pool.*

General equilibrium Because the wage is fixed and households have linear utility, each industry's profit landscape depends only on its own number of active firms. Under *idiosyncratic shocks and no inter-industry input-output linkages or demand spillovers*, the multi-industry economy with I industries each characterized by (φ_i, μ_i) has industries that evolve independently, with aggregate spectral density a superposition of industry-specific peaks at frequencies $\omega_i = \sqrt{\varphi_i/\mu_i}$. The isolation of frequency from the shock process is therefore a *cross-sectional* prediction in this scope: different industries cycle at different frequencies determined by their respective market structures. Under common shocks, IO linkages, or demand spillovers, the spectral superposition does

not survive aggregation, and the clean cross-industry decomposition becomes an approximate one; Section 6 discusses robustness under CRRA preferences, and Online Appendix U documents that the empirical cross-industry frequency dispersion lies outside the envelope of a multi-sector RBC with input-output linkages and AR(p) sectoral shocks.

Exogenous exit at rate δ is standard (Hopenhayn, 1992). It ensures a steady flow of agents returning to the search pool, maintaining the timing search problem at every date.

Let $n(t)$ denote the mass of active production units at date t and $S(t) = M - n(t)$ the mass of searchers. Throughout the paper, I interpret n as the mass of active *establishments*—physical production units such as plants, stores, or branches—rather than firms (legal entities). The establishment interpretation matches both empirical series used in Sections 5–5.4: BEA investment in structures measures the rate of new establishment construction, and BDS establishment entry and exit rates measure establishment births and deaths. The matching-congestion microfoundation of μ (Section 2.3) is also most natural at the establishment level, where bottlenecks in permits, specialist labor, equipment, and sites constrain rapid adjustment. The aggregate state of the economy is fully characterized by $n(t)$ and its rate of change $\dot{n}(t)$.

2.2 Matching efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency. But beyond a point, additional firms congest the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = zl_i^{\gamma} \cdot m(n)$. Profit is $\pi(n) = \bar{\pi}(w) \cdot m(n)$, where $\bar{\pi}$ is the standard profit given the wage.

Assumption 2 (Gaussian scale efficiency)**.**
Scale efficiency takes the Gaussian form:

$$m(n) = \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \varphi > 0,\ n_p > 0. \tag{2}$$

Interpretation of Assumption 2. The function peaks at $n = n_p$ (optimal market scale) and falls off symmetrically. The parameter φ governs the sharpness of the trade-

off: high φ means a narrow peak (sharp trade-off); low φ means a broad peak (robust to deviations). Online Appendix B provides a matching-theoretic microfoundation.

For the analytical results, I work with the reduced-form Gaussian profit:

$$\pi(n) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \pi_0 > 0. \tag{3}$$

The semi-elasticity is exactly linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{4}$$

This linearity delivers exact harmonic oscillations (Theorem 1). The general case—where wage congestion adds a non-quadratic correction to $\ln \pi$—produces anharmonic oscillations with the same leading-order frequency (Proposition 3).

2.3 The timing friction: congestion in the entry process

Entry is mediated by a matching function. The $S(t) = M - n(t)$ searchers compete for R units of *entry resources*—permits, specialized labor, venture capital, factory sites—to form matches:

Assumption 3 (Entry matching)**.** $m(S, R) = S^\alpha R^{1-\alpha}$, $\alpha \in (0, 1)$.

The matching infrastructure is tuned to clear the steady-state throughput $e_{ss} = \delta n_p$. Deviations in the realized replacement flow from e_{ss}—in either direction—strain the infrastructure: rapid entry overloads permitting and construction; net contraction (replacement flow falling below attrition) overloads resale and reorganization. Both reduce the matching probability per searcher and produce the timing friction.

Proposition 1 (Micro-foundation of the timing friction)**.**
Under Assumptions 1–3, the entry matching friction produces, to second order around the steady state, the timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ *with*

$$\mu = \frac{1 - \alpha}{\alpha\,(\delta n_p)^2}. \tag{5}$$

Sketch. A symmetric bottleneck-strain penalty $g(\dot{n}^2/(\delta n_p)^2)$ with $g(0) = 0$ and $g'(0) = 1/2$ multiplies the Cobb-Douglas matching probability; we adopt the specific functional form $g(z) = \ln(1 + z/2)$ (yielding $\xi(\dot{n}) = [1 + \dot{n}^2/(2(\delta n_p)^2)]^{-(1-\alpha)/\alpha}$) as a representative

element of this equivalence class. Taylor expansion at second order gives $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$ with $\mu = (1-\alpha)/[\alpha(\delta n_p)^2] > 0$. *The symmetry of g in $\dot{n}$ is an additional knife-edge condition* beyond Assumption 3: under asymmetric upside (permitting/construction) and downside (resale/reorganization) strain, with second-order curvatures $\mu_+ \neq \mu_-$, the leading-order ironing condition would carry a cubic term in $\dot{n}$ that produces amplitude-dependent frequency, of the same general form analyzed in Proposition 5. The exact-harmonic result of Theorem 1 requires the symmetric specification; the qualitative cycle survives asymmetric strain. Online Appendix G provides the detailed derivation. ■

The timing friction μ decreases in the matching elasticity α, decreases in the steady-state turnover δn_p, and increases in scarcity of entry resources $1/R$. Substituting (5) into $\omega = \sqrt{\varphi/\mu}$ gives the structural frequency

$$\omega = \delta n_p \sqrt{\alpha\varphi/(1-\alpha)}, \tag{6}$$

connecting cycle frequency to three independently measurable objects: matching elasticity α (estimated in the range $[0.5, 0.7]$ for labor markets, Petrongolo and Pissarides 2001), profit curvature φ, and steady-state turnover δn_p. Effective output is $y(t) = \pi(n(t))\,\xi(\dot{n}(t))$. The quadratic form in $\dot{n}$ arises generically from any smooth symmetric matching friction; Online Appendix C derives the same ξ from discrete-time proportional adjustment costs.

2.4 Combined structure

Combining the Gaussian profit and the Gaussian friction, effective output is:

$$y(t) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n-n_p)^2 - \frac{\mu}{2}\dot{n}^2\Big). \tag{7}$$

The effective output is a bivariate Gaussian kernel in $(n, \dot{n})$, centered at $(n_p, 0)$. Both terms impose quadratic penalization of deviation from the optimum: n_p for market scale and $\dot{n} = 0$ for adjustment speed.

2.5 Infrastructure supplier and the Rayleigh friction χ

The matching technology of Section 2.3 treats R, the stock of entry resources, as fixed. In practice, $R(t)$ is supplied by a separate sector—permitting authorities, construction labor, specialist equipment manufacturers—that produces these resources at a flow cost convex in the rate of provision. The bottleneck-strain damping coefficient χ that selects the stationary cycle amplitude (Proposition 7) and carries the welfare result (Theorem 2) emerges from this supplier's first-order condition. We make the supplier explicit so that χ is a derived equilibrium object, distinct from the per-firm sunk entry cost κ and from the matching friction μ.

Setup An infrastructure supplier produces a flow $R(t)$ of entry resources at instantaneous resource cost

$$C(R(t)) = c_0 + c_1\, R(t) + \tfrac{1}{2}\, c_2 \left(R(t) - R_{ss}\right)^2, \qquad c_2 > 0, \tag{8}$$

where R_{ss} is the steady-state level of resource provision tuned to clear $e_{ss} = \delta n_p$, and $c_0, c_1, c_2 \geq 0$ are smoothness primitives describing fixed, marginal, and convex congestion costs of resource provision. The supplier sells $R(t)$ to the matching technology at the competitive price equal to its marginal cost, $p_R(t) = c_1 + c_2(R(t) - R_{ss})$.

Resource demand from the entry process The entry flow $\dot{n} + \delta n$ requires a resource quantity that varies one-for-one with the deviation of flow from steady state. Let $R(t) = R_{ss} + r\left(\dot{n} - 0\right)$ for some reduced-form linkage coefficient $r > 0$ that maps a unit deviation in flow to a unit deviation in resource demand.[1] Substituting into (8), the deviation of the supplier's cost flow from its steady-state level is

$$C(R(t)) - C(R_{ss}) - c_1 r\, \dot{n}(t) = \tfrac{1}{2}\, c_2\, r^2\, \dot{n}(t)^2. \tag{9}$$

The constant $C(R_{ss})$ scales the household budget but is not informative for cycle dynamics; the linear term $c_1 r \dot{n}$ averages to zero over a cycle (since $\int \dot{n}\, dt = 0$ on a closed orbit) and contributes only to the steady-state ironing constant. The *quadratic*

[1] The linear linkage is the local approximation around the steady state; the structure of the matching technology in Section 2.3 pins r to the inverse of the steady-state matching elasticity, $r = (\delta n_p)/\alpha$, but the qualitative argument here does not depend on this specific value.

term, by contrast, is strictly positive whenever $\dot{n} \neq 0$ and represents a real output deduction borne by households who consume the supplier's output.

Definition of χ and the welfare margin Defining the Rayleigh friction coefficient as

$$\boxed{\chi \equiv c_2\, r^2,} \tag{10}$$

the supplier's quadratic cost flow (9) equals $\frac{1}{2}\chi\,\dot{n}^2$.[2] The household's flow budget constraint absorbs this deduction: aggregate consumption equals firms' aggregate profit flow $\pi(n)\xi(\dot{n})$ *net* of the supplier's quadratic resource cost,

$$c(t) = \pi(n(t))\,\xi(\dot{n}(t))\,n(t) - \tfrac{1}{2}\chi\,\dot{n}(t)^2 + \text{constants}. \tag{11}$$

The welfare cost of cycles in Section 6.2 then follows from time-averaging (11) along the stationary distribution of $(n, \dot{n})$, with the equipartition identity (Online Appendix I) yielding $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$.

Why χ is distinct from κ and μ The three coefficients map to three different microeconomic primitives. The per-firm sunk entry cost κ is paid by individual entrants at the moment of entry and enters the free-entry condition $V(t) = V^{\text{search}}(t) + \kappa$ (Definition 1(ii)); it pins the steady-state ironing level but does not appear in the equation of motion or the welfare formula. The matching friction $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ governs the curvature of the multiplicative kernel $\xi(\dot{n}) = e^{-\mu\dot{n}^2/2}$ inside firms' profit flow and pins the cycle frequency $\omega = \sqrt{\varphi/\mu}$. The Rayleigh friction $\chi = c_2 r^2$ governs the supplier's quadratic resource-cost flow and pins the cycle amplitude through the welfare margin. Whether a real-world entry-cost intervention reduces κ (a per-firm transfer that does not damp cycles), reduces μ (a matching-technology improvement that speeds cycles), or reduces χ (a supplier-side capacity expansion that amplifies cycles by reducing damping) is an institutional question we revisit in Section 6.3.

The supplier microfoundation makes the welfare loop transparent. The $\chi\dot{n}^2$ flow is paid by households as an output deduction at the welfare margin, not by individual entrants as part of κ, so it does not appear in the firm-level ironing condition and

[2] The Rayleigh friction enters the equilibrium equation of motion through $-\partial\mathcal{R}/\partial\dot{n}$ where $\mathcal{R} = \frac{1}{2}\chi\dot{n}^2$ is the Rayleigh dissipation function (Online Appendix I); the rate of energy outflow is $2\mathcal{R} = \chi\dot{n}^2$, equal to the supplier's resource flow (9) at the welfare margin. The factor of one-half between the dissipation function $\mathcal{R}$ and the dissipation rate $2\mathcal{R}$ is the standard Rayleigh convention.

does not double-count. Online Appendix I provides the full derivation, including the supplier's first-order condition, the household budget that absorbs the resource flow, and the proof that χ enters the equation of motion as a Rayleigh damping coefficient via the energy-balance argument summarized above.

3 Ironing Along Time: Equilibrium and the Main Result

The equilibrium condition that agents cannot profit by retiming their entry—the ironing condition—pins a rate of entry along any approach to optimal scale. That pinned rate becomes momentum the industry cannot discard at the optimum without creating arbitrage. The result is cycles, and in the frictionless limit they are exactly harmonic with frequency $\omega = \sqrt{\varphi/\mu}$. This section formalizes the argument.

3.1 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning effective flow payoff $\pi(n(t))\xi(\dot{n}(t))$ per unit time, discounted at rate $r \geq 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, \pi(n(t))\, \xi(\dot{n}(t))\, \mathrm{d}t. \tag{12}$$

The agent maximizes $V(\tau)$ over τ, taking the aggregate path $\{n(t)\}$ as given. An agent who does not enter at date t remains in the search pool, with continuation value

$$V^{\text{search}}(t) = \max_{\tau \geq t} e^{-r(\tau-t)}\big[V(\tau) - \kappa\big], \tag{13}$$

where $\kappa \geq 0$ is the per-firm sunk entry cost paid at the moment of entry. The entry-matching infrastructure (permits, construction crews, specialist services) is also bottlenecked, with strain rising in the squared deviation of flow from the steady-state rate; this strain is summarized at the macroscopic level by a Rayleigh friction coefficient $\chi > 0$, which produces a damping force $-\chi\dot{n}$ in the equilibrium equation of motion (Online Appendix I). Two parameters are therefore distinct in the model: κ is the dollar amount the marginal entrant pays at entry; χ is the macroscopic

damping coefficient governing the bottleneck-strain mechanism. The constant and linear parts of the entry cost (including κ) pin the ironing level but do not dissipate; the Rayleigh friction χ generates the damping $\eta = \chi/\mu$ that appears in the operative equilibrium. We treat χ as a reduced-form amplitude-selection primitive rather than as a separately accounted output flow: it enters the equilibrium through the damping force and the welfare identity (28), but is not double-counted as a contemporaneous resource deduction at the welfare margin. We track κ and χ separately throughout: κ enters the free-entry condition above, while χ enters the damped equation of motion and the welfare formula.

3.2 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date. Together with the initial condition $n(0)$, it determines the entire path of the economy through the aggregate consistency condition $\dot{n} = e - \delta n$.

Definition 1 (Timing search equilibrium)**.**
A timing search equilibrium *is a path* $\{n(t)\}_{t\geq 0}$ *and an entry policy* $e(\cdot)$ *such that:*

(i) ***Individual optimality****: Each agent chooses her entry date to maximize* $V(\tau)$*.*

(ii) ***Temporal free entry (with complementary slackness)****:* $V(t) \leq V^{search}(t) + \kappa$ *at all dates, with equality whenever* $e(t) > 0$ *and* $n(t) < M$*. At dates where the population cap binds* ($n(t) = M$) *or the entry flow is zero* ($e(t) = 0$)*, the inequality may be strict.*

(iii) ***No profitable timing deviation****:* $V(\tau) \leq V(t)$ *for all* τ *and all* t *in the support of entry.*

(iv) ***Aggregate consistency****:* $\dot{n}(t) = e(t) - \delta\, n(t)$*, where* $e(t) \geq 0$*.*

(v) ***Feasibility****:* $n(t) \leq M$ *for all* t*.*

The no-profitable-timing-deviation condition Condition (iii) is the key requirement. In Burdett and Mortensen (1998), the analogous condition is that no firm can profit by reposting its wage, which yields equal profits across wages. Here, the condition that no agent can profit by retiming her entry yields equal profits across

dates. If any date offered strictly higher value, agents would crowd that date, changing $n(t)$ until the advantage was eliminated.

Commitment versus dynamic optimization The formulation in (12)–(13) is a commitment problem: agents choose τ taking the entire future path $\{n(t)\}$ as given. An alternative formulation would have agents observe the current state $(n(t), \dot{n}(t))$ at each instant and decide dynamically whether to enter. The ironing condition makes the two equivalent. Since $V(\tau) = V^*$ for all τ in equilibrium, an agent who could re-optimize at any date t would find no profitable deviation—she is indifferent across all entry dates. The commitment equilibrium is therefore also an equilibrium of the dynamic problem: agents who can freely re-time their entry choose not to.

3.3 The ironing condition

The key equilibrium condition is that agents cannot profit by retiming their entry. This is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998): Burdett-Mortensen equalize profits across wages so that no firm gains by reposting, and timing search equalizes payoffs across dates so that no agent gains by retiming. I call this the *ironing condition*, since it "irons out" any temporal variation in the value of entry.

Proposition 2 (Ironing condition)**.**
In a timing search equilibrium where entry occurs at all dates, agents are indifferent across entry dates. Equivalently, the effective flow payoff is equalized:

$$\pi(n(t))\,\xi(\dot{n}(t)) = C \qquad \textit{for all } t, \tag{14}$$

for some constant $C > 0$.

Proof. Suppose entry occurs at all dates. For any two dates τ_1, τ_2 in the support of entry, condition (iii) of Definition 1 requires $V(\tau_1) \leq V(\tau_2)$ and $V(\tau_2) \leq V(\tau_1)$, so $V(\tau_1) = V(\tau_2)$. Since the support is all of $\mathbb{R}_+$, $V(\tau) = V^*$ for all $\tau \geq 0$. The equilibrium path $n(\cdot)$ is C^1 by aggregate consistency (condition (iv)) and the smoothness of π and

ξ, so $V(\tau)$ is differentiable in τ.[3] Differentiating (12):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

Setting $\frac{\mathrm{d}V}{\mathrm{d}\tau} = 0$ (since V is constant): $\pi(n(\tau))\xi(\dot{n}(\tau)) = (r+\delta)V^* \equiv C$. ■

Regularity condition The condition that entry occurs at all dates is satisfied whenever the searcher pool $S(t) = M - n(t)$ is bounded away from zero along the equilibrium path and the timing friction μ is positive. With a large population M and moderate cycle amplitude $A < M - n_p$, the condition holds. Online Appendix F provides a formal verification.

3.4 The main result

The equilibrium under Gaussian profits and no dissipation is an exact harmonic oscillator with frequency $\omega = \sqrt{\varphi/\mu}$ (Theorem 1). This subsection derives that result. The frictionless case is not the most realistic specification, but it is the analytically tractable one: the ironing condition reduces to a conservative harmonic oscillator whose orbit and period admit closed-form expressions at every amplitude.

The closed-form frequency carries over to the full operative equilibrium with Rayleigh friction and shocks, which is a damped, stochastically forced version of the same oscillator:

$$\ddot{n} + \eta\,\dot{n} + \frac{\varphi}{\mu}\,(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t, \tag{15}$$

with $\eta = \chi/\mu$ the damping rate from convex congestion in entry processing (with χ the Rayleigh friction coefficient introduced above) and σ_ε the shock volatility. Section 4 characterizes this damped-stochastic equilibrium and the stationary amplitude it selects. All three cases—frictionless, damped without shocks, damped with shocks—share the same natural frequency ω and differ only in whether the oscillator is isolated, decaying, or continuously excited.

Taking logarithms of the ironing condition (14) and substituting the functional

[3] The differentiability of V in τ requires that $t \mapsto \pi(n(t))\xi(\dot{n}(t))$ is locally integrable along the equilibrium path, which holds whenever $n(\cdot) \in C^1$. This regularity is guaranteed by the smoothness of π (Assumption 2) and ξ (Section 2.3), combined with the continuity of the entry flow $e(\cdot)$ in condition (iv).

forms (7):

$$-\frac{\varphi}{2}(n-n_p)^2 - \frac{\mu}{2}\dot{n}^2 = E, \tag{16}$$

where $E = \ln C - \ln \pi_0 < 0$ is constant along the equilibrium path. Differentiating with respect to t and dividing by $\dot{n} \neq 0$:

$$\ddot{n} = -\frac{\varphi}{\mu}(n-n_p). \tag{17}$$

Theorem 1 (Exact harmonic oscillations)**.**
In a timing search equilibrium (Definition 1) under Assumptions 1 and 2, the equilibrium path is:

$$n(t) = n_p + A\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}. \tag{18}$$

The solution is exact and global; no linearization is required. The amplitude A and phase ϕ are determined by initial conditions, and the ironing condition holds with $E = -\frac{\varphi}{2}A^2$.

Proof. The ironing condition (Proposition 2) gives (16). Differentiation yields the harmonic oscillator equation (17), whose general solution is (18) with $\omega = \sqrt{\varphi/\mu}$. The solution is exact because the Gaussian specification makes the semi-elasticity $f(n)$ exactly linear (equation (4)), so no Taylor approximation is needed. Substituting back: $\frac{\varphi}{2}A^2\cos^2(\omega t+\phi) + \frac{\mu}{2}\omega^2 A^2 \sin^2(\omega t+\phi) = \frac{\varphi}{2}A^2 = -E$. ■

Verification, scope, and selection The construction establishes existence: for any amplitude $A \leq \delta n_p/\sqrt{\omega^2+\delta^2}$ (the binding sufficient condition for $e(t) \geq 0$ throughout the cycle, equivalent to $\delta(n_p - A) \geq \omega A$), the pair $(n(t), e(t))$ is a timing search equilibrium, with the entry policy $e(t) = \dot{n}(t) + \delta n(t)$ non-negative and feasibility holding for $M > n_p + A$ (Online Appendix J). The condition is tighter than the looser $S(t) > 0$ requirement that searchers be available, and it sharpens the empirical-applicability scope: industries whose net establishment entry rate routinely goes substantially negative (i.e., $|\dot{n}| > \delta n_p/2$) are outside the regime where the ironing condition holds globally, and the equilibrium of Proposition 2 should be read as applying to its complement. The center structure leaves the amplitude A free; the stochastic extension (Proposition 6) and the dissipation result (Proposition 7) resolve this multiplicity by selecting a unique stationary distribution from primitives. The model is deliberately one-sided: exit occurs through exogenous attrition at rate δ,

so the equilibrium is a theory of activation timing and replacement pressure rather than of endogenous exit. The empirical exercises use the net establishment entry rate (entry minus exit) as the model-consistent counterpart of $\dot{n}$; combining timing search with Jovanovic (1982)-style learning is a natural extension that would endogenize exit (Section 6).

Remark 1 (Isochrony)**.**
The frequency ω does not depend on the amplitude A. Small and large business cycles have the same frequency. The economic reason is that both the speed of adjustment and the distance to travel scale with the size of the displacement. A large shock creates strong congestion pressure, driving rapid correction—but the economy must also traverse a longer path (overshooting proportionally far past n_p before returning). A small shock creates weaker pressure and slower adjustment, but the path is shorter. Since speed and distance both scale linearly with A, the time per cycle $T = 2\pi\sqrt{\mu/\varphi}$ is independent of severity. Isochrony extends to the damped equilibrium: successive swings preserve their period even as the envelope decays (Figure 1).

Structural decomposition The frequency $\omega = \sqrt{\varphi/\mu}$ decomposes into two primitives. The numerator φ is the curvature of the profit landscape: how sharply profits fall when the industry deviates from optimal scale. The denominator μ is the timing friction: how costly it is to adjust the number of active firms rapidly, arising from congestion in the entry matching process. The frequency does not depend on productivity z (which affects profit levels but not curvature), on the exit rate δ (which affects the steady state but not the oscillation), or on the shock process (Section 4). Table 1 contrasts this decomposition with the way standard frameworks pin cycle frequency.

3.5 Properties of the cycle

Corollary 1 (Properties)**.**
Under the conditions of Theorem 1:

(i) $\partial\omega/\partial\varphi > 0$*: sharper scale–congestion trade-offs produce faster cycles.*

(ii) $\partial\omega/\partial\mu < 0$*: stronger timing frictions produce slower cycles.*

(iii) ω *is independent of productivity* z*, of the shock process* σ_ε*, and of the cycle amplitude* A*.*

Table 1: What determines cycle frequency?

	Frequency determined by	**Cross-industr**
RBC	Shock process and internal propagation (capital, labor, IO linkages)	Industry-specifi
NK	Policy rule	Common across
Beaudry et al. (2020)	Bifurcation condition	Not addressed
Timing search	$\boldsymbol{\omega = \sqrt{\varphi/\mu}}$	**Industry-spec**

Notes: In RBC and NK models, the spectral peak reflects the shock process or the policy rule—objects common across industries. In Beaudry et al. (2020), the frequency emerges from a Hopf bifurcation that depends on the full parameterization. In timing search, the frequency is a closed-form function of two market-structure primitives (φ: profit-landscape curvature; μ: timing friction), is exact under linear utility, and is isolated from the shock process (Proposition 6).

(iv) The period is $T = 2\pi\sqrt{\mu/\varphi}$.

(v) ***Frequency-welfare separation****: the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ depends only on the market-structure primitives (φ, μ) and is independent of the shock process σ_ε and the Rayleigh friction coefficient χ. The welfare cost (Section 6.2) depends only on the orthogonal pair $(\sigma_\varepsilon, \chi)$. Cycle severity is the intermediate object: with $\eta = \chi/\mu$ the matching friction μ cancels in $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, so severity depends on $(\sigma_\varepsilon, \chi, \varphi)$. Frequency and welfare cost are therefore governed by disjoint sets of primitives.*

(vi) The amplitude A is free: the system is a center. In the stochastic extension (Section 4), the stationary amplitude distribution is determined by the balance of shocks and the profit-landscape curvature.

(vii) All orbits are exactly symmetric. Asymmetric cycles arise under general hump-shaped π (Proposition 5).

Proof. Parts (i)–(iv) follow from direct differentiation of $\omega = \sqrt{\varphi/\mu}$. Part (v) is Theorem 2. Part (vi) holds because the ironing condition (16) admits a one-parameter family of solutions indexed by E (equivalently, by $A = \sqrt{-2E/\varphi}$). Part (vii) follows from the exact cosine form of (18). ■

3.6 Phases of the cycle

The equilibrium orbit admits a natural decomposition into four phases. When $n < n_p$ and $\dot{n} > 0$, the scale externality dominates: profits rise with the number of firms, attracting more entrants and accelerating entry (boom). Once n exceeds n_p, congestion sets in and profits fall, but the entry wave cannot stop instantly because of matching frictions—the economy overshoots. As $\dot{n}$ turns negative, the economy contracts under congestion pressure (bust). Finally, n falls below n_p and the scale externality again favors entry, but exit momentum carries the economy below the optimum (undershoot) before the cycle repeats.

3.7 Role of hump-shaped profits

With pure congestion ($\varphi = 0$ or $f(n) < 0$ everywhere), profits fall monotonically in the number of firms and the economy converges to the steady state without oscillation, as in Hopenhayn (1992). Oscillations require all three ingredients. The scale externality makes booms self-reinforcing: more firms raise profits, attracting further entry. Congestion reverses booms: too many firms lower profits, discouraging entry. The timing friction prevents instantaneous correction: matching congestion means entry waves take time to build and time to unwind, generating the overshooting that sustains the cycle. Removing any one eliminates the oscillation.

Remark 2 (Why the ironing condition produces oscillation)**.** *The ironing condition* (16) *requires that profit-landscape loss and adjustment cost sum to a constant along the equilibrium path. When the economy is near optimal scale ($n \approx n_p$), the profit-landscape term is small, so the adjustment-cost term must be large: the economy is adjusting rapidly. When the economy is far from optimal scale, profit-landscape losses are large, so adjustment must be slow. This creates overshooting: as the economy approaches n_p, it has built up rapid adjustment that matching frictions prevent it from shedding instantly, carrying it past the optimum.*

Mathematically, the constant-sum structure of (16) *is identical to energy conservation in classical mechanics (with profit-landscape loss playing the role of potential energy and adjustment cost the role of kinetic energy). This structure guarantees closed orbits—the economy must return to its starting point rather than spiral inward or outward—and explains why the ironing condition produces periodic oscillation rather than monotone convergence.*

3.8 Beyond the Gaussian: general profits

The Gaussian profit function delivers exact results, but the mechanism does not depend on it. This subsection shows that the frequency formula extends to general hump-shaped profits, quantifies the approximation error, and derives a new prediction about cycle asymmetry.

Proposition 3 (General hump-shaped profits)**.**
Under any C^2 profit function $\pi(n)$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, and a timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$, the ironing condition

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E$$

admits periodic orbits in the potential $U(n) = -\frac{1}{\mu}\ln \pi(n)$, which captures how the profit landscape confines the economy near optimal scale. The frequency near n_p is:

$$\omega = \sqrt{\frac{|f'(n_p)|}{\mu}} + O(A^2), \tag{19}$$

where $f'(n_p) = \pi''(n_p)/\pi(n_p) < 0$.

Proof. See Online Appendix E. ■

For the Gaussian, $|f'(n_p)| = \varphi$ exactly. For $\pi(n) = n^\alpha(a - bn)$, $|f'(n_p)| = \alpha/n_p^2 + b^2/(a - bn_p)^2$. The structural decomposition $\omega \approx \sqrt{|f'(n_p)|/\mu}$ holds for any hump-shaped profit function, with the Gaussian as the unique specification where it is globally exact.

A natural concern is that the exactness is a knife-edge property. The following result characterizes the correction.

Proposition 4 (Frequency correction under asymmetric profits)**.**
For a C^4 profit function with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, the oscillation frequency at amplitude A is:

$$\omega(A) = \omega_0\big(1 + c_4\, A^2 + O(A^4)\big), \tag{20}$$

where $\omega_0 = \sqrt{|f'(n_p)|/\mu}$ and c_4 depends on $U^{(3)}(n_p)$ and $U^{(4)}(n_p)$, with $U(n) = -\frac{1}{\mu}\ln \pi(n)$ the effective potential. For the Gaussian, $c_4 = 0$ exactly (isochrony).

Proof. See Online Appendix H. ■

A numerical check across 8 smooth exponential-family hump-shaped profit specifications shows that the Gaussian-based prediction $T_0 = 2\pi\sqrt{\mu/\varphi}$ stays within 1.6% of the exact orbit-integrated period at $A/n_p = 0.30$ (Online Appendix S). The Gaussian is therefore not the necessary specification but the analytically tractable benchmark; the closed-form frequency is robust to general hump-shaped profits at empirically relevant amplitudes.

The Gaussian also produces exactly symmetric cycles, which is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Relaxing symmetry turns this weakness into a prediction.

Proposition 5 (Cycle asymmetry)**.**
When $\pi'''(n_p) \neq 0$, the periodic orbit is asymmetric. The leading-order asymmetry parameter—the fractional difference in time spent above versus below n_p—is:

$$\mathcal{A} \equiv \frac{t_{above} - t_{below}}{t_{below}} = \frac{\pi'''(n_p)}{4\,|\pi''(n_p)|}\,A + O(A^2). \tag{21}$$

When $\pi'''(n_p) < 0$ (profit falls faster on the congestion side), the economy spends more time below n_p: slow expansions and sharp contractions.

Proof. See Online Appendix H. ■

The degree of asymmetry is pinned by the ratio $\pi'''(n_p)/|\pi''(n_p)|$, which is independently measurable from the curvature and skewness of industry profit functions. Online Appendix S illustrates the asymmetric orbit numerically and verifies the anharmonic frequency correction (Proposition 4) against simulations under quartic and skewed profit perturbations.

4 Stochastic Extension

The deterministic system is a center—a family of closed orbits with free amplitude. Small stochastic perturbations reveal the resonance of this center; the damped extension (Section 4.3, Proposition 7) then selects a unique stationary distribution by combining shocks with Rayleigh damping.

4.1 Adding shocks

Consider an i.i.d. disturbance to profits:

$$\tilde{\pi}(n,t) = \pi(n) + \sigma_\varepsilon \, \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} (0,1). \tag{22}$$

The ironing condition no longer holds exactly: E drifts stochastically according to $\mathrm{d}E \approx \frac{\sigma_\varepsilon \, \varepsilon_t}{\pi(n)} \dot{n} \, \mathrm{d}t$, and the economy migrates between orbits.

4.2 Properties

Proposition 6 (Stochastic limit cycle)**.**
Under small shocks (22):

(i) The spectral density of $n(t)$ peaks at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ in the undamped small-shock limit; with Rayleigh damping $\eta = \chi/\mu > 0$, the observed PSD peak is $\omega_{PSD} = \sqrt{\varphi/\mu - \eta^2/2}$, equal to ω_0 up to an $O(\eta^2)$ correction (Proposition 7(iv)).

(ii) The peak frequency is determined by (φ, μ) at leading order and by the damping ratio η at second order, not by σ_ε.

(iii) Even i.i.d. shocks produce persistent cycles: the persistence is endogenous to the equilibrium structure.

Proof sketch. Write the stochastic system in action–angle coordinates (A, ϕ), where $n = n_p + A\cos\phi$ and $\dot{n} = -A\omega_0 \sin\phi$. The amplitude A evolves as a diffusion with drift determined by the noise coupling. The spectral density of $n(t)$ is the power spectrum of a stochastically forced harmonic oscillator; its transfer function has a resonance peak at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ in the undamped limit and at $\omega_{\text{PSD}} = \sqrt{\varphi/\mu - \eta^2/2}$ under Rayleigh damping, independent of σ_ε in either case. The full derivation is in Online Appendix D. ■

Role of shock structure The structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ is a property of the market structure—the profit-landscape curvature and the matching friction—not of the shock process or of agents' behavior. It is invariant to the shock variance, the shock persistence, and the equilibrium amplitude. The observed spectral density is the

product of the transfer function $|H(\nu)|^2$, which peaks at $\omega_{\text{PSD}} = \sqrt{\omega_0^2 - \eta^2/2}$, and the shock spectrum $S_\varepsilon(\nu)$. With i.i.d. shocks, S_ε is flat, so the observed peak coincides with ω_{PSD}, which equals ω_0 up to the small damping correction $O(\eta^2)$ (Proposition 7(iv)); shock variance σ_ε^2 scales the peak height without shifting its location. With persistent shocks, $S_\varepsilon(\nu)$ is tilted toward low frequencies (a persistent process concentrates power in slow fluctuations). The product $|H(\nu)|^2 S_\varepsilon(\nu)$ then peaks slightly below ω_{PSD}, because the tilted shock spectrum gives more weight to the left side of the transfer-function peak. This is an observational displacement, not a change in the structural frequency: ω_0 is unchanged, but an econometrician estimating the spectral peak from data would find it slightly below ω_0. The displacement is negligible whenever the transfer function is sharp relative to the shock persistence.

This matches the central finding of Beaudry et al. (2020): persistent boom–bust dynamics from short-lived disturbances. In Beaudry et al. (2020), the endogenous persistence comes from financial-friction complementarities; here, it comes from the scale–congestion trade-off that traps the economy in persistent oscillations.

A numerical simulation of the stochastic system $(n, \dot{n})$ via Euler–Maruyama discretization confirms the predicted spectral concentration: the simulated PSD peak coincides with $\omega_0 = \sqrt{\varphi/\mu}$ up to the small $O(\eta^2)$ damping shift, while an AR(1) benchmark produces a featureless monotonically decaying spectrum. Online Appendix K reports the simulation parameters, the spectral density figure, additional phase-portrait results, and BDS nonlinearity tests.

4.3 The operative equilibrium: damped stochastic oscillator

In the operative equilibrium—with Rayleigh bottleneck dissipation and profit shocks—the industry behaves as a damped, stochastically forced oscillator at the same natural frequency $\omega = \sqrt{\varphi/\mu}$. Dissipation from convex congestion in the entry-replacement pipeline pulls the industry toward optimal scale, shocks re-excite it, and the stationary distribution of $(n - n_p)$ has variance $\sigma_\varepsilon^2/(2\eta\varphi\mu)$ (Proposition 7). Theorem 1's closed ellipse is the frictionless limit ($\eta \to 0$, $\sigma_\varepsilon \to 0$); the operative equilibrium is the economically relevant extension.

The entry-replacement bottleneck consumes real resources—permits, specialized equipment, site development, reorganization services, write-downs—at a rate that is convex in $\dot{n}$ around steady state (Online Appendix I). The convex part of this

cost, with Rayleigh friction coefficient χ at steady-state flow, dissipates output at a rate $\chi\dot{n}^2$, while shocks with standard deviation σ_ε continuously perturb the industry. Together, these two forces turn the closed orbit of Theorem 1 into the damped stochastic oscillator (15). Online Appendix I derives the damped equation in two steps. First, differentiating the ironing condition gives the conservative Newton equation $\mu\ddot{n} + \varphi(n - n_p) = 0$. Second, balancing the rate at which conservative energy is lost against the bottleneck's output-dissipation rate $\chi\dot{n}^2$ yields the additional drag term $\chi\dot{n}$, with signs fixed by the requirement that the bottleneck consumes rather than produces output:

$$\ddot{n} + \eta\,\dot{n} + \frac{\varphi}{\mu}(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t, \tag{23}$$

with $\eta = \chi/\mu$ emerging endogenously from the ratio of the Rayleigh friction coefficient χ and the matching friction μ of Proposition 1. The natural frequency $\omega = \sqrt{\varphi/\mu}$ is unchanged: damping and shocks shape *when* the industry visibly oscillates and *how severely*, but they leave the industry's structural pitch untouched.

Figure 1 illustrates the impulse-response behaviour of (23) in the deterministic-but-damped case ($\sigma_\varepsilon = 0$, $\eta > 0$, $x(0) = A_0 \neq 0$), which is the case that the empirical echo analysis in Section 5.4 examines in manufacturing investment. Three different damping rates η are plotted on the same axes, with fixed (φ, μ) so that the natural frequency is common across the three trajectories. Each trajectory's envelope decays exponentially at rate $\eta/2$, but the zero crossings occur at nearly identical times: the pitch ω is preserved while the volume fades. This decoupling of declining amplitude from preserved period is a distinctive feature of the timing-search equilibrium and contrasts with linear models such as AR(1) in which the decay rate and the period are mechanically linked.

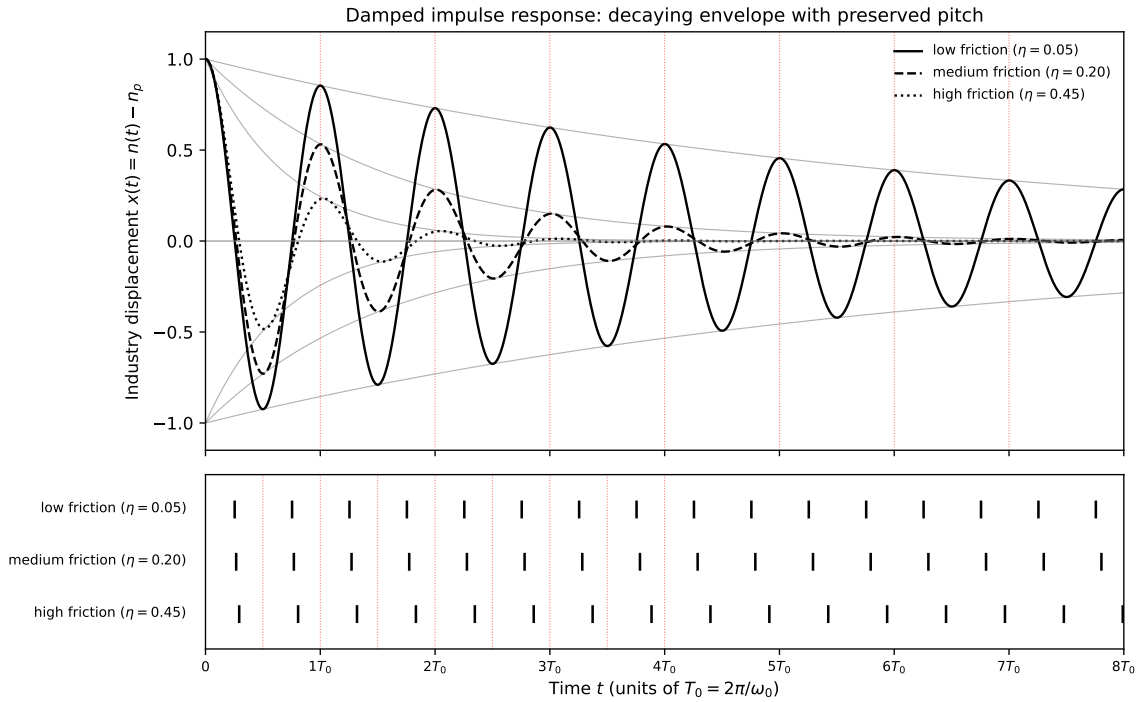
Proposition 7 (Amplitude selection)**.**
With Rayleigh friction coefficient $\chi > 0$ (damping $\eta = \chi/\mu > 0$) and shocks $\sigma_\varepsilon > 0$:

(i) Without shocks ($\sigma_\varepsilon = 0$): orbits spiral inward to $(n_p, 0)$ with decay half-life $\ln 2 \cdot 2\mu/\chi$.

(ii) With shocks: the system has a unique stationary distribution with variance:

$$\mathbb{E}\big[(n - n_p)^2\big] = \frac{\sigma_\varepsilon^2}{2\,\eta\,\varphi\,\mu} = \frac{\sigma_\varepsilon^2}{2\,\chi\,\varphi}, \tag{24}$$

Figure 1: Damped impulse response of the timing-search equilibrium



Notes: The damped harmonic oscillator $\ddot{x} + \eta\,\dot{x} + (\varphi/\mu)\,x = 0$ with $x(0) = A_0$, $\dot{x}(0) = 0$, and fixed natural frequency $\omega_0 = \sqrt{\varphi/\mu} = 1$. Three damping rates $\eta \in \{0.05, 0.20, 0.45\}$ are plotted. Top panel: trajectories with the exponential envelope $A_0 e^{-\eta t/2}$ shown in grey; the dotted red grid marks integer multiples of $T_0 = 2\pi/\omega_0$. Bottom panel: zero crossings of each trajectory, showing that the half-period marker (red) coincides with each crossing. Higher friction damps the envelope faster but leaves the pitch ω_0 essentially unchanged. The empirically relevant signature—decaying amplitude with preserved period—is documented in the manufacturing investment echo of Section 5.4.

where the second equality uses $\eta = \chi/\mu$, so that the matching friction μ cancels.

(iii) The Hilbert envelope $A = \sqrt{(n - n_p)^2 + (\dot{n}/\omega_0)^2}$ follows a Rayleigh distribution with scale $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, so that $\mathbb{E}[A^2] = 2\sigma_x^2 = \sigma_\varepsilon^2/(\chi\varphi)$.

(iv) The spectral peak frequency of the driven oscillator's stationary spectrum is $\omega_{peak} = \sqrt{\varphi/\mu - \eta^2/2}$, obtained by maximizing the transfer function $|H(\omega)|^2 = 1/[(\omega_0^2 - \omega^2)^2 + \eta^2\omega^2]$ over ω. (The homogeneous damped natural frequency $\sqrt{\varphi/\mu - \eta^2/4}$, at which the unforced solution oscillates, is a separate object.) For small η, both quantities are negligibly below $\sqrt{\varphi/\mu}$.

Proof. See Online Appendix I. ■

Economic interpretation Substituting $\eta = \chi/\mu$ into the stationary standard deviation gives $\sigma_x = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu} = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, in which the matching friction μ cancels exactly. Industries with volatile profits (σ_ε large), low Rayleigh friction (χ small), or flat profit landscapes (φ small) experience larger cycles. Industries whose entry pipelines have stiffer Rayleigh friction—pharmaceuticals, electric power, mining—dissipate more energy per unit excess flow, producing smaller fluctuations relative to the shocks they face. The severity formula depends on three observable quantities: shock volatility (from profit time series), the Rayleigh friction coefficient (from establishment-level data on the response of entry-flow speed to shocks), and profit curvature (from the cross-section). The matching friction μ governs only the cycle frequency, not the severity. The model now explains both the frequency and the severity of business cycles from primitives.

Combining with the welfare formula (28), where the welfare loss includes both the displacement-curvature loss $(\varphi/2)\mathbb{E}[x^2]$ and the friction-output loss $(\mu/2)\mathbb{E}[\dot{n}^2]$ (equal by equipartition):

$$\mathbb{E}[\mathcal{L}] \approx \frac{\varphi}{2}\,\mathbb{E}[A^2] = \varphi\,\mathbb{E}\big[(n-n_p)^2\big] = \frac{\sigma_\varepsilon^2}{2\,\eta\,\mu} = \frac{\sigma_\varepsilon^2}{2\,\chi}. \tag{25}$$

Higher Rayleigh friction χ *reduces* welfare loss by damping the cycle—a "stabilization through friction" result. The magnitude of cycles is pinned by shock volatility, the Rayleigh friction coefficient, and profit curvature $(\sigma_\varepsilon, \chi, \varphi)$; together with the welfare identity in (25), the welfare cost reduces further to $(\sigma_\varepsilon, \chi)$ as φ cancels. Online Appendix K illustrates the four dissipation/shock regimes via phase portraits and verifies the stationary Rayleigh distribution numerically against simulations.

4.4 Frequency-welfare separation

The closed-form structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ (Theorem 1), the stationary variance $\sigma_x^2 = \sigma_\varepsilon^2/(2\chi\varphi)$ (Proposition 7 after substituting $\eta = \chi/\mu$, so that μ cancels), and the welfare cost $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ (Section 6.2) together establish the paper's central structural claim: *structural frequency and welfare cost have disjoint determinants*, with cycle severity standing as an intermediate observable depending on the overlapping set $(\sigma_\varepsilon, \chi, \varphi)$. We state this as a single theorem.

Theorem 2 (Frequency-welfare separation)**.**

In the timing search equilibrium under Assumptions 1–3 and the damped-stochastic specification of Proposition 7, with $\eta = \chi/\mu$:

$$\begin{aligned} \textit{structural frequency:} \quad & \omega_0 = \sqrt{\varphi/\mu} && \textit{depends only on } (\varphi, \mu); \\ \textit{cycle severity:} \quad & \sigma_x = \sigma_\varepsilon/\sqrt{2\,\chi\,\varphi} && \textit{depends on } (\sigma_\varepsilon, \chi, \varphi); \\ \textit{welfare cost:} \quad & \mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi) && \textit{depends only on } (\sigma_\varepsilon, \chi). \end{aligned}$$

The determinant sets of the structural frequency $\{\varphi, \mu\}$ and the welfare cost $\{\sigma_\varepsilon, \chi\}$ are disjoint. The structural frequency is invariant to the shock volatility σ_ε, the Rayleigh friction coefficient χ, the shock persistence (in the i.i.d. specification), and the realized cycle amplitude A; the welfare cost is invariant to the market-structure primitives (φ, μ) that govern timing and to the per-firm entry cost κ. Cycle severity is the intermediate observable: with $\eta = \chi/\mu$ the matching friction μ cancels, so severity depends on $(\sigma_\varepsilon, \chi, \varphi)$, overlapping with both determinant sets and isolated from neither. The observed PSD peak frequency inherits the structural frequency up to an $O(\eta^2)$ correction: $\omega_{PSD} = \sqrt{\varphi/\mu - \eta^2/2}$ (Proposition 7(iv)), so empirically measured spectral peaks reflect (φ, μ) at leading order with a small damping-dependent shift.

Proof. The structural-frequency claim is Theorem 1 (deterministic linearized dynamics admit eigenfrequency $\omega_0 = \sqrt{\varphi/\mu}$, depending on (φ, μ) alone). The severity claim follows by substituting $\eta = \chi/\mu$ into Proposition 7(ii): $\sigma_x^2 = \sigma_\varepsilon^2/(2\eta\varphi\mu) = \sigma_\varepsilon^2/(2\chi\varphi)$, so μ cancels exactly. The welfare claim follows from substituting the same identity and $\mathbb{E}[(n - n_p)^2] = \sigma_\varepsilon^2/(2\chi\varphi)$ into (28); the dependence on φ also cancels, leaving $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$. The observed-PSD-peak claim is Proposition 7(iv). ■ ■

The theorem has three immediate methodological consequences. First, cross-industry variation in ω_0 identifies (φ, μ) alone in the equilibrium of the model, with no contamination from differences in shock processes, the Rayleigh friction coefficient χ, or the per-firm entry cost κ (the $O(\eta^2)$ shift to the observed PSD peak is small at the empirical damping levels of Section 5, $\eta^2/(2\omega_0^2) \lesssim 5\%$). The cross-sectional regression of Section 5.3 therefore tests a structural prediction even though the proxies for φ are imperfect. Second, two industries with identical market structure (φ, μ) but different shock environments cycle at the same frequency in the model. This makes the model's cross-industry frequency prediction structural rather than a reduced-form artifact of common shocks propagating through industry-specific filters. Third, industrial policy

that targets μ (entry regulation) or φ (cluster development) affects timing alone in the model; only policies that move σ_ε or χ (the Rayleigh friction coefficient) affect welfare. The policy implications in Section 6.3 follow directly from this separation.

Scope of the separation: leakage under alternative welfare metrics Theorem 2's welfare claim relies on three conditions: (i) linear utility, (ii) the welfare metric in equation (28), and (iii) treating the supplier's $\chi\dot{n}^2$ flow as paid at the welfare margin (Section 2.5). Relaxing each of these reintroduces some leakage of (φ, μ) into the welfare cost. We document the leakage numerically in Online Appendix T and summarize here.

Under CRRA utility with risk-aversion coefficient $\sigma_{\text{ra}} \in \{1, 2, 5\}$, the consumption-equivalent welfare loss departs from the linear-utility baseline $\sigma_\varepsilon^2/(2\chi)$ by +8% to +11% at the median across a representative grid of $(\varphi, \mu, \chi, \sigma_\varepsilon)$, with maximum leakage of about ±100% in the worst cells (high φ or low χ, where amplitude is large and risk aversion bites the consumption density most). Under an alternative welfare metric that includes the supplier's resource flow as an additional welfare-relevant deduction (rather than at the welfare margin), the welfare cost becomes $\mathbb{E}[\mathcal{L}^{\text{alt}}] = \sigma_\varepsilon^2/(2\chi) + \sigma_\varepsilon^2/(2\mu)$, which explicitly reintroduces μ. The median leakage is +20% across the grid, reaching +80% at low μ. We therefore state the separation as exact under conditions (i)–(iii), with median leakage of order 10–25% under standard CRRA preferences or under alternative welfare metrics that double-count the supplier's resource flow. The separation is exact, not approximate, in the welfare framework adopted here, but is not robust to the specific welfare metric chosen, and we make this caveat explicit.

4.5 Comparison to Beaudry et al. (2020)

The closest existing model of endogenous business cycles is Beaudry et al. (2020), who generate a stochastic limit cycle from financial frictions via Hopf bifurcation. Both models share the abstract architecture of a stochastic limit cycle—a destabilizing force, a bounding force, and small i.i.d. shocks that maintain the cycle—but the economic content differs: BGP place financial frictions in the destabilizing role and diminishing returns in the bounding role, with amplitude selected through the Hopf bifurcation; timing search places agglomeration externalities in the destabilizing role and congestion in the bounding role, with amplitude selected through Rayleigh bottleneck dissipation.

The two frameworks also differ methodologically in how the cycle frequency object is represented. In BGP, the cycle frequency emerges implicitly from a Hopf bifurcation condition that involves the full parameterization of the financial-friction model; it must be read off numerically once the full system is calibrated, and is not industry-decomposable in a useful sense. In timing search, the equilibrium frequency is the closed form $\omega_0 = \sqrt{\varphi/\mu}$, where φ and μ are each cross-sectionally measurable. The cross-industry regression in Section 5.3 is feasible *because* of this closed-form structure: the predicted slope of $\ln \omega$ on $\ln \varphi$ is exactly $1/2$, and on $\ln \mu$ exactly $-1/2$, derived from the analytical formula rather than from a numerical bifurcation calibration. The two mechanisms are nevertheless complementary rather than competing: financial frictions and market-structure frictions could coexist, potentially generating models with multiple spectral peaks. Online Appendix Q provides the side-by-side mechanism table.

Model horse-race A side-by-side simulation comparison of timing search against an AR(1) RBC benchmark, a New Keynesian AR(2) with common Taylor-rule propagation, and a Van der Pol limit cycle à la Beaudry et al. (2020) confirms that none of these alternatives jointly delivers the three motivating empirical patterns—cross-industry frequency heterogeneity, post-shock periodicity at industry-specific frequencies, and overshoot above pre-shock trend—in the parametrizations considered. Only timing search produces all three. Online Appendix O details the simulations and reports the binary scorecard and figure.

Identification: multi-sector RBC envelope under input-output linkages and AR(p) shocks A more demanding alternative is a multi-sector RBC with input-output linkages and industry-specific AR(p) productivity shocks. Such a framework can produce industry-specific spectral peaks that cluster by supply-chain linkage, raising the question whether the cross-industry frequency dispersion documented in Section 5.3 reflects timing-search heterogeneity in (φ, μ) or correlated propagation through input-output structure. We address this by simulating a 16-sector log-linear environment $y_{i,t} = (W y_{t-1})_i + a_{i,t}$ with sector-specific AR(2) productivity $a_{i,t} = \rho_{1i}\, a_{i,t-1} + \rho_{2i}\, a_{i,t-2} + \sigma_i\, \varepsilon_{i,t}$ and an input-output matrix W, extracting the spectral peak of each simulated $y_{i,t}$ via the same Burg-AR procedure used on the data, and comparing the cross-industry standard deviation of $\ln \omega_i$ to the empirical magnitude.

We sweep across three IO schemes (no IO, diagonal-dominant own-share, sparse cross-linkages) and two AR-coefficient distributions (homogeneous, heterogeneous), with 100 replications per cell of the 3×2 design. The 95% envelope of simulated $\sigma(\ln \omega_i)$ ranges from $[0.24, 0.41]$ under homogeneous AR coefficients and $[0.29, 0.49]$ under heterogeneous AR coefficients, across all three IO schemes. The observed manufacturing dispersion is $\sigma(\ln \omega_i) = 0.68$, and 0% of simulated replications reach this magnitude. The empirical cross-industry spectral dispersion is therefore outside the multi-sector RBC envelope under reasonable calibrations: correlated sectoral shock propagation through input-output structure does not, in this exercise, reproduce the observed dispersion. Online Appendix U reports the full envelope across IO schemes and AR specifications.

5 Quantitative Framework

The model delivers three testable cross-industry predictions. First, cycle frequency should vary with market-structure proxies, not with the shock process. Second, large one-time shocks should trigger deterministic oscillations at each industry's structural frequency, not at a common frequency inherited from the shock. Third, industries displaced by a common shock should overshoot their pre-shock trend rather than converge monotonically. Section 5.3 tests the first using the Ellison-Glaeser concentration index across 16 manufacturing 3-digit industries, with a parallel cross-section across 10 BDS supersectors using independently measured spatial concentration as a robustness extension. Section 5.4 tests the third prediction using establishment entry-exit data across 19 BDS sectors, complemented by a qualitative departure from AR(1) dynamics in the negative-autocorrelation pattern of net entry rates and by the post-COVID overshoot. Section 5.4 tests the second prediction using Fisher's g-test on manufacturing and oil investment after the Depression/WWII and 1979 oil shocks.

5.1 Cross-industry predictions

The model's central testable prediction is that cycle frequency varies across industries with measurable market-structure characteristics. Taking logs of $\omega_i = \sqrt{\varphi_i/\mu_i}$ yields

a regression framework:

$$\ln \omega_i = \frac{1}{2} \ln \varphi_i - \frac{1}{2} \ln \mu_i + u_i, \tag{26}$$

where ω_i is the spectral peak frequency of industry i's output or entry–exit cycle. The coefficients $\frac{1}{2}$ on $\ln \varphi$ and $-\frac{1}{2}$ on $\ln \mu$ are *exact* predictions of the model—not free parameters—so equation (26) is testable with industry-level proxies for scale-externality sharpness and adjustment friction.

The qualitative predictions are as follows:

(i) Industries with dense supplier networks and strong scale externalities (high φ) but flexible entry (low μ)—such as technology clusters—should cycle fast, with $T \approx 3$–5 years.

(ii) Manufacturing industries with moderate linkages and moderate frictions should cycle at intermediate frequencies, with $T \approx 5$–10 years.

(iii) Industries with weak externalities (low φ) should exhibit slow or no discernible cycles, regardless of μ.

(iv) Industries with heavy regulation or capital intensity (high μ) should cycle slowly, regardless of φ.

Crucially, two industries with identical shock processes but different (φ, μ) will cycle at different frequencies. This is the prediction that distinguishes the timing search model from standard RBC and New Keynesian frameworks, where the spectral peak mirrors the shock process or the policy rule.

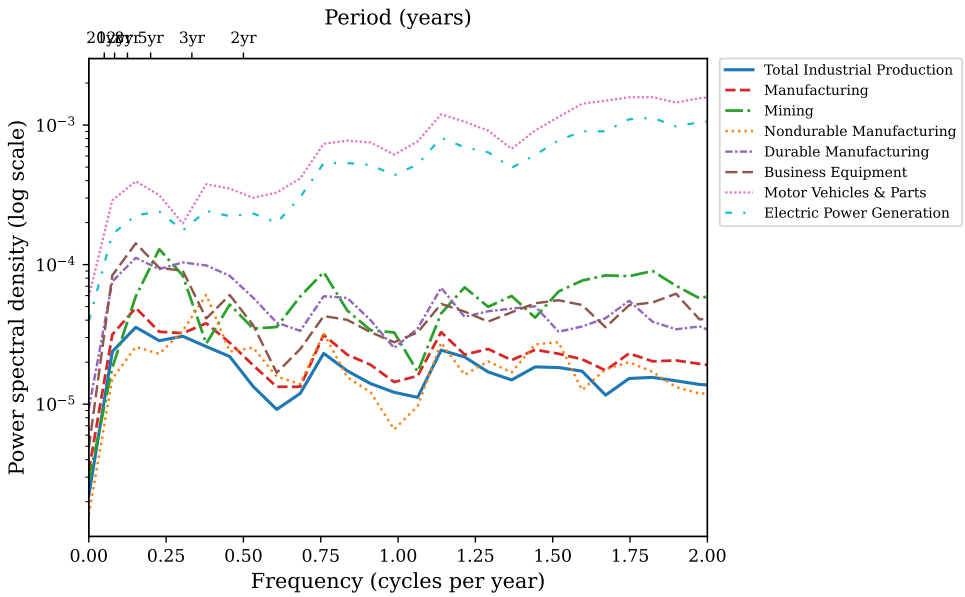
5.2 Spectral evidence from U.S. industry data

I evaluate the cross-industry prediction using monthly industrial production indices from FRED for eight U.S. sectors over 1972–2024. The series span a range of market structures: from Motor Vehicles & Parts (dense supplier networks, strong scale externalities) to Mining (dispersed extraction, weak inter-firm linkages) to Electric Power Generation (regulated, capital-intensive).

Figure 2 plots the power spectral density of log-differenced production for each industry. Two features stand out. First, spectral peaks vary substantially across

industries. Motor Vehicles, Durable Manufacturing, Business Equipment, and aggregate Manufacturing concentrate spectral mass near periods of 6–7 years. Mining peaks at 4.4 years. Nondurable Manufacturing and Electric Power peak near 2.6 years. Second, the cross-industry clustering is informative: industries sharing supply-chain linkages—durables, motor vehicles, business equipment—cluster at similar frequencies despite producing very different goods. This is consistent with ω depending on market structure, not on product-specific shocks.

Figure 2: Spectral density of U.S. industry production growth

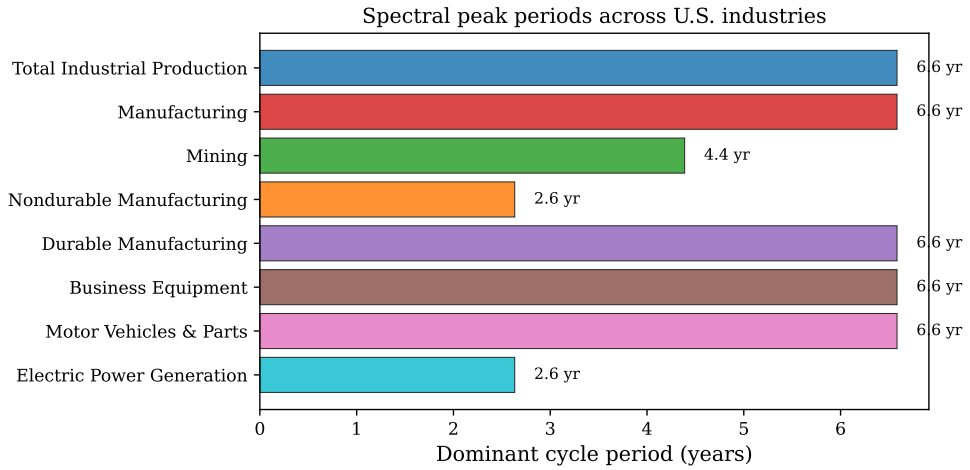


Notes: Power spectral density (Welch's method) of monthly log-differenced industrial production indices from FRED, 1972:M1–2024:M12. The top axis labels convert frequency (cycles per year) to period (years). Industries with strong supply-chain linkages (Motor Vehicles, Business Equipment, Durable Manufacturing) cluster near $T \approx 6$–7 years, while industries with different market structures (Mining, Nondurables, Electric Power) show distinct peak frequencies.

Figure 3 summarizes the dominant cycle period for each industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). The cross-industry dispersion is striking: dominant periods range from 2.6 years (Electric Power, Nondurable Manufacturing) to 6.6 years (Durable Manufacturing, Motor Vehicles). This two-and-a-half-fold variation in cycle frequency across industries that share a common macroeconomic environment—the same monetary policy, the same aggregate demand shocks, the same financial conditions—is difficult to reconcile with models that derive spectral properties from a common

shock process. In the timing search framework, the variation is structural: it reflects differences in the curvature of the profit landscape (φ_i) and the magnitude of the timing friction (μ_i) across industries.

Figure 3: Cross-industry variation in spectral peak periods



Notes: Dominant cycle period (years) for each U.S. industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). Industries with strong scale–congestion externalities (Motor Vehicles, Durable Manufacturing, Business Equipment) cluster near $T \approx 6$–7 years. Industries with weaker or different externality structures (Mining, Nondurables, Electric Power) cycle at distinct frequencies.

Consistency check: nonlinear dependence in industry dynamics As a consistency check rather than a discriminating test, we apply the BDS test (Brock et al., 1996) to AR(1)-filtered industry residuals; the i.i.d. null is rejected at the 1% level for all eight industries with z-statistics ranging from 16.2 (Durable Manufacturing) to 21.8 (Motor Vehicles & Parts). This establishes that industry dynamics contain nonlinear or nonstationary structure beyond AR(1), which is consistent with the timing-search mechanism but is also consistent with GARCH, regime switching, and structural breaks (the 1972–2024 sample spans the Great Moderation and COVID, both of which can produce comparable BDS rejection magnitudes). The BDS exercise therefore confirms departure from linear AR(1) dynamics but does not discriminate among the nonlinear alternatives. Full results across embedding dimensions 2–6, with AR(1) and AR(p)+GARCH pre-filtering specifications, are in Online Appendix K.

5.3 Cross-industry regression with external proxies

Motivating broad-sector pattern (N=7) A baseline regression on seven broad industry groups, using BEA proxies for the two model primitives—inverse I-O self-purchase ratio for φ and capital per employee for μ—yields the predicted signs and an R^2 of 0.75, with $\hat{\beta}_\varphi = 1.31$ ($p = 0.027$) and $\hat{\beta}_\mu = -0.19$. The univariate regression on $\ln \hat{\varphi}_i$ alone gives slope 0.85 ($R^2 = 0.65$, $p = 0.028$). Online Appendix P documents the proxy construction, multivariate table, and leave-one-out robustness. We treat this result as a motivating broad-sector pattern rather than a stand-alone confirmatory finding—the $N = 7$ sample is sensitive to industry definitions and to the specific BEA tables used to construct the proxies. The headline empirical test of the paper is the $N = 16$ three-digit specification with the Ellison and Glaeser (1997a) concentration index, reported below.

Headline regression: N=16 three-digit manufacturing To test whether the cross-industry relationship survives disaggregation, we expand to $N = 16$ three-digit manufacturing industries (NAICS 311–339 where data is available). Two adjustments to the specification are required. First, we replace Welch's method with Burg's AR spectral estimation (order 24), which produces continuous spectral peak frequencies and eliminates the bin-discretization that arises at this resolution.[4] Second, we replace the input–output and capital-intensity proxies with the Ellison and Glaeser (1997a) geographic concentration index γ_i. The motivation for the proxy swap is empirical: at the 3-digit level, the I-O and capital-intensity proxies do not preserve the broad-sector relationship (the multivariate $N = 16$ specification with these proxies yields wrong-signed coefficients with $R^2 = 0.16$; details in Online Appendix P). The BEA detailed I-O table aggregates from 6-digit to broad sectors using non-uniform weights; capital intensity within manufacturing also varies far less than across the broader $N = 7$ sample. The Ellison and Glaeser (1997a) index, by contrast, is constructed at the 3-digit manufacturing level by design and provides a direct measure of agglomeration externalities.

Table 2 reports the results. In the univariate specification, the Ellison and Glaeser (1997a) concentration index significantly predicts cycle frequency at the 5% level

[4] With 636 monthly observations and Welch segments of 256 points, the frequency resolution is $1/256 \approx 0.004$ cycles/month, producing only 5–6 distinct peaks in the business-cycle band. AR spectral estimation has no such constraint.

($\hat{\beta}_\gamma = 0.20$, HC1 SE 0.07, $p_{\beta=0} = 0.013$, $R^2 = 0.39$, $N = 16$). The sign matches the theory: geographically concentrated industries—those with stronger agglomeration externalities (higher φ)—cycle faster. The point estimate, however, rejects the structural prediction $\beta_\gamma = 1/2$: the t-statistic against the null $\beta_\gamma = 1/2$ is -4.29 ($p_{\beta=1/2} = 0.0008$). We therefore read the headline regression as a *sign test* of the directional prediction (cycle frequency rises in geographic concentration) rather than as a structural test of the magnitude.

Reliability-ratio interpretation The gap between $\hat{\beta}_\gamma = 0.20$ and the structural 0.50 corresponds in classical errors-in-variables terms to a reliability ratio of $\lambda = \hat{\beta}_\gamma/0.5 = 0.40$, i.e., 60% of the cross-industry variance in $\ln \gamma_i$ would have to be measurement noise relative to the latent $\ln \varphi_i$ for the OLS slope to recover 1/2. Sampling noise in the Ellison and Glaeser (1997a) index alone is small at this level: under the EG sampling-error formula $\mathrm{Var}(\hat{\gamma}_i) \approx 2\gamma_i^2/N_i$ with plant counts $N_i \in [10^3, 10^5]$ for 3-digit manufacturing, the implied $\mathrm{Var}(\ln \hat{\gamma}_i)$ averages to about 4×10^{-4} against an observed cross-industry variance of 0.74, a reliability ratio of $\lambda_{\text{sampling}} \approx 0.9995$. The sampling-noise EIV-corrected slope is therefore essentially unchanged at $\hat{\beta}_\gamma^{\text{EIV}} = 0.20$, with the rejection of $\beta = 1/2$ preserved at the same significance ($t = -4.28$). Closing the gap to 1/2 would require *specification noise*—the conceptual mismatch between geographic concentration γ and the structural curvature φ—to account for 60% of the variance in $\ln \gamma$, an upper bound that is empirically unverifiable without an instrument and which we do not defend. We therefore prefer the cleaner reading: the cross-section confirms the model's directional prediction and rejects its magnitude prediction; cross-industry variation in geographic concentration is a useful but imperfect proxy for the profit-landscape curvature object the model identifies.

Adding the structural timing-friction proxy $\mu_i^{\text{struct}} = 1/(\delta_i n_{p,i})^2$ derived from Census Business Dynamics Statistics (Proposition 1) leaves R^2 unchanged at 0.39, and the additional regressor is not informative in the multivariate specification. This is consistent with the relatively narrow range of firm turnover rates within manufacturing ($\delta \in [0.065, 0.095]$), which compresses the variation in μ_i^{struct}.

The μ channel is not identified at headline resolution Two external proxies for μ with real cross-industry variation—capital-stock-to-investment ratios from BEA Fixed Assets tables (a replacement-horizon proxy) and order-of-magnitude estimates of

plant-level capital-adjustment quadratic-cost coefficients from Cooper and Haltiwanger (2006)—both yield positive point estimates ($\hat{\beta}_{\mu,A} = 0.67$, HC1 SE 0.93, $p_{\beta=0} = 0.49$; $\hat{\beta}_{\mu,B} = 0.67$, HC1 SE 0.59, $p_{\beta=0} = 0.29$ at $N = 10$ shared-sample manufacturing), with the wrong sign relative to the structural prediction $\beta_\mu = -1/2$. Neither rejects the zero null. The two proxies are highly correlated with each other ($r = 0.99$), so they are capturing a common construct; the construct simply does not predict cycle frequency in the predicted direction at the data's resolution. We therefore state plainly that the empirical evidence in this paper identifies the φ channel only. The μ channel prediction $\beta_\mu = -1/2$ from the closed-form $\omega = \sqrt{\varphi/\mu}$ is structural but is not directly tested. A credible cross-industry test of μ would require an external proxy with both real variation and a defensible mapping to the matching-friction object $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$; we leave this for future work and note the limitation in the abstract and conclusion. Online Appendix V documents the proxy construction and full regression results.

Table 2: Structural regression: geographic concentration and cycle frequency

Coefficient	Estimate	HC1 Std. Error	Theory
$\hat{\alpha}$ (intercept)	1.36	(0.54)	—
$\hat{\beta}_\gamma$ (E-G concentration)	0.20	(0.07)	0.50
R^2		0.39	
N		16	
p-value		0.010	

Notes: OLS regression of $\ln \omega_i$ (AR spectral peak frequency, business-cycle band) on $\ln \gamma_i$ (Ellison and Glaeser 1997a geographic concentration index). Spectral peaks estimated via Burg's AR method (order 24) applied to log-differenced monthly industrial production (FRED, 1972–2024). The sample consists of 16 three-digit NAICS manufacturing industries. HC1 heteroskedasticity-robust standard errors (White, 1980). The estimated slope rejects $\beta_\gamma = 0$ at the 5% level ($t = 2.86$) and rejects the structural prediction $\beta_\gamma = 1/2$ at the 0.1% level ($t = -4.29$). The cross-section therefore confirms the directional prediction (sign test) but rejects the magnitude prediction.

Expanding from $N = 7$ to $N = 16$, the R^2 falls from 0.75 to 0.39 and the coefficient point estimate falls from 1.31 to 0.20. The directional sign and statistical significance against the zero null are preserved at the higher resolution; the magnitude moves from above the structural prediction (1.31) to below (0.20), with both samples rejecting $\beta = 1/2$ in different directions. We read this as evidence that the proxies $\hat{\varphi}$ and γ each capture parts of the latent profit-landscape curvature with different attenuation

properties, not as a structural identification of the coefficient.

A natural check at finer industry resolution—$N = 86$ four-digit NAICS manufacturing industries using BDS `vcn4` data—collapses to $\hat{\beta}_\gamma = 0.027$ ($p = 0.705$, $R^2 = 0.002$). The null is expected mechanically: Ellison et al. (2010) publish γ at the three-digit level only, so every four-digit child inherits its parent's γ, producing vertical columns in the scatter and driving the slope to zero. The four-digit regression therefore tests the *inheritance* of γ rather than the predictive power of geographic concentration at four-digit resolution; the three-digit level is the appropriate statistical resolution for E-G-published data. Online Appendix R provides details.

A second cross-section beyond manufacturing (N=10) The headline N=16 regression uses the FRED industrial production series and the published Ellison et al. (2010) index, both manufacturing-only. As a robustness check across data sources, we assemble a parallel cross-section across NAICS supersectors covered by BDS (1978–2022). The cycle frequency is the spectral peak of the detrended net establishment entry rate (Burg AR(6) estimation); the concentration measure is $G_i = \sum_s (z_{s,i} - x_s)^2$, the raw E-G component computed from BDS state-by-sector establishment data.[5] Restricting to the ten supersectors with a clear interior spectral peak in the business-cycle band (Fisher's g-test $p < 0.10$, period strictly below the 20-year estimator cap), the regression $\ln \omega_i = \alpha + \beta \ln G_i + \varepsilon_i$ yields $\hat{\beta} = 0.107$ (HC1 SE 0.043, $p_{\beta=0} = 0.037$, $R^2 = 0.39$). The same sign and a similar R^2 emerging from a different data source for ω (BDS net entry rather than FRED industrial production), a different concentration measure (BDS-derived G rather than Ellison et al. (2010) γ), and a non-manufacturing-heavy sector mix indicates that the headline cross-sectional relationship is not an artifact of the manufacturing 3-digit sample. As with the manufacturing headline, the slope rejects the structural $\beta = 1/2$ ($t = -9.19$, $p_{\beta=1/2} < 0.001$); we read this regression as a sign-test confirmation that does not pin down the structural magnitude. The implied critical reliability ratio for the 2-digit aggregation is $\lambda_{\text{crit}} = 0.21$, consistent with the larger attenuation expected at coarser industry resolution.

[5] G_i is the raw component of the Ellison-Glaeser γ, well-defined for non-manufacturing sectors where the published Ellison et al. (2010) indices are unavailable. Aggregation to the 2-digit supersector level attenuates the absolute magnitude of G relative to the 3-digit γ values, because within-supersector variation in geographic concentration averages out.

Identification The model has eight parameters: φ (profit curvature), μ (timing friction), χ (Rayleigh friction coefficient), κ (per-firm sunk entry cost), n_p (optimal scale), π_0 (profit level), δ (exit rate), and σ_ε (shock volatility). Among these, the pair (φ, μ) governs the structural cycle frequency. The ratio φ/μ is identified directly from the spectral peak: $\varphi/\mu \approx (2\pi/T^*)^2$ at leading order, with an $O(\eta^2)$ correction $\omega^2_{\text{PSD}} = \varphi/\mu - \eta^2/2$. The peak *height* scales with the shock variance σ^2_ε (linear transfer-function). The peak *width*, normalized by frequency, is governed by the damping ratio $\eta = \chi/\mu$: stronger Rayleigh friction broadens the resonance, while shock variance scales the height without changing the normalized width. The individual values of φ and μ require external information: φ from the Ellison and Glaeser (1997b) concentration index or input–output linkage measures, and μ from the speed of entry–exit responses to demand shocks (Dunne et al., 1988) or capital adjustment cost estimates (Cooper and Haltiwanger, 2006). The Rayleigh friction χ is identified from the stationary cycle severity given $(\sigma_\varepsilon, \varphi)$ via $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$. The per-firm cost κ is identified separately from the steady-state ironing level (the offset of the value of waiting from the value of immediate entry).

5.4 Additional diagnostics

The cross-sectional regression in Section 5.3 is the paper's structural test, identifying the φ channel through the geographic-concentration proxy. We complement it with three additional diagnostics that are consistent with the timing-search mechanism but rest on weaker identification: oscillatory features in BDS net entry rates that lie outside the AR(1) class, post-shock periodicity in two historical investment series at industry-specific structural frequencies, and an aggregate post-COVID overshoot pattern. We present these as suggestive rather than definitive evidence.

Oscillatory entry dynamics across industries (BDS, 1978–2022) For each of 19 NAICS sectors, we compute the linearly detrended net establishment entry rate (entry rate minus exit rate) and examine two AR(1)-distinguishing features. First, four sectors (Construction, Wholesale, Retail, Real Estate) exhibit significantly negative sample autocorrelation at lag 4 outside the 95% Bartlett band. The population ACF of any scalar AR(1) is ρ^k, which is non-negative at even lags; the four-sector pattern is therefore inconsistent with AR(1) at the level of these sectors, although a single negative sample ACF across many sectors and many lags could arise under

multiple-testing without familywise correction. Second, Fisher's g-test (Fisher, 1929) applied to the detrended series after AR pre-whitening rejects its (i.i.d. Gaussian) null at the 5% level in 10 of the 19 sectors, with spectral peaks concentrated near $T \approx 8$–9 years; the test should be read as a periodogram-irregularity test rather than as a deterministic-periodicity test. The sign-changing autocorrelation pattern at the level of net entry is also consistent with vintage-capital replacement at the cohort margin, which produces alternation in net entry through cohort exit timing; the two mechanisms are not directly distinguishable from the ACF moment alone, and we discuss alternative discriminating moments below. Online Appendix M provides the full methodology, sector-by-sector results, and figure.

Post-shock periodicity in historical impulse responses A second diagnostic uses BEA investment-in-structures data (1930–2024) to test whether large one-time shocks trigger oscillations at industry-specific structural frequencies. Manufacturing structures investment after the Great Depression/WWII oscillates at $T = 5.0$ years (Fisher's g-test $p = 0.036$ in the 1934–1973 echo window; $p = 0.578$ in the 1974–2013 non-echo window), while mining/oil exploration after the 1979 crisis oscillates at $T = 3.25$ years ($p = 0.016$). *The three "manufacturing" period estimates in this paper refer to overlapping but not identical objects:* (i) $T = 5.0$ years for manufacturing structures investment in the 1934–1973 echo window (BEA Fixed Investment by Industry, dollar-weighted); (ii) $T \approx 6$–7 years for the spectral peaks of monthly industrial production at the 3-digit NAICS manufacturing level over 1972–2024 (FRED IP, output-weighted); and (iii) $T \approx 8$–9 years for the spectral peaks of detrended net establishment entry rates over 1978–2022 (BDS, establishment-count-weighted). The three estimates differ because they measure different aggregates (dollar investment, output, establishment counts) over different sample windows; we report all three honestly rather than reconciling them to a single number. The detected echo period is invariant to ± 3-year window shifts; only marginal p-values vary at the boundaries. The cross-industry frequency difference is consistent with different (φ, μ) across industries and harder to reconcile with explanations that would produce a common frequency. The vintage-capital alternative for manufacturing (Kydland and Prescott, 1982b) is compatible with rather than competing with the timing-search interpretation: replacing wartime capital is itself a wave of establishment-level construction subject to the same timing frictions. Online Appendix L provides the full data construction,

methodology, robustness across windows, and figures.

The post-COVID natural experiment The COVID-19 pandemic displaced all U.S. industries below trend in 2020 simultaneously. Timing search predicts that displaced industries should not return monotonically to trend but rather overshoot the pre-shock level: the ironing condition forces the economy past optimal scale before cycling back. In the BDS data, 14 of 19 NAICS sectors (74%) overshoot above their pre-COVID linear trend in 2021 or 2022, against an AR(1)-monotone-convergence baseline of approximately 30%. The 30% baseline is constructed by simulating, for each BDS sector, an AR(1) process calibrated to that sector's pre-COVID (1978–2019) net entry rate residual standard deviation and persistence, applying the same 2020 displacement, and computing the fraction of simulated sample paths that exceed the pre-COVID linear trend in any year of 2021–2022 under *monotone* mean reversion. A binomial test of the observed 74% against the 30% null rejects at $p < 0.001$. We acknowledge that this binomial test exploits only the sign and timing of post-COVID dynamics; a phase-content test (peak amplitude and timing of the post-2020 industry response at the estimated ω_i) would carry more structural information but requires more data than the 2020–2024 window provides for stable estimation. *The structural cross-sectional prediction that overshoot magnitude correlates with sector concentration is not borne out:* the variation in observed overshoot is dominated by sector-specific COVID exposure (lockdown intensity, demand shifts, supply-chain disruption), violating the common-shock idealization at the sector level. The COVID diagnostic therefore carries directional information about the aggregate overshoot pattern but no cross-sectional information about the structural channel; a clean cross-sectional natural-experiment test of the structural channel requires a shock more uniform across sectors than COVID was.

6 Discussion

The closed-form frequency $\omega = \sqrt{\varphi/\mu}$ and its separation from the welfare cost of cycles rest on five maintained assumptions: a hump-shaped profit landscape, the Gaussian specification, linear utility, a convex matching friction, and exogenous exit. This section addresses each in turn, showing that the cycle mechanism survives relaxations while the closed-form exactness is Gaussian-specific. It then computes the welfare cost

of the cycles and the implications for industrial policy, where the frequency-welfare separation delivers concrete predictions for CHIPS-Act-style entry subsidies.

6.1 Modeling choices and limitations

The Gaussian specification The Gaussian profit function (3) produces exactly symmetric cycles: booms and busts have equal duration. This is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Under general hump-shaped profit functions (Proposition 3), cycles become asymmetric, with the degree of asymmetry determined by the skewness of $\pi(n)$ around n_p. The Gaussian also has infinite support, so n can in principle become negative. In practice, for $A \ll n_p$, this is quantitatively irrelevant.

Amplitude selection In the deterministic model without dissipation, the amplitude A is a free parameter: the ironing condition $E = -\frac{\varphi}{2}A^2$ admits a one-parameter family of orbits. Section 4.3 resolves this through convex-congestion dissipation (Proposition 7). With Rayleigh friction coefficient $\chi > 0$ and shocks $\sigma_\varepsilon > 0$, the balance of dissipation and shocks selects a unique stationary distribution with $\sigma_x = \sqrt{\mathbb{E}[(n-n_p)^2]} = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, depending on shock volatility, the Rayleigh friction coefficient, and profit curvature—all three observable. The structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ is invariant throughout this process (the observed PSD peak shifts negligibly to $\sqrt{\varphi/\mu - \eta^2/2}$ for small η), so amplitude selection operates without altering the cycle frequency at leading order.

The timing friction The timing friction is micro-founded in Section 2.3 from congestion in the entry matching process, yielding $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ (Proposition 1). Cross-industry variation in μ arises from three structural sources: (i) variation in α (matching technology differs across industries); (ii) variation in δn_p (industries with higher turnover have smaller μ); (iii) variation in R (industries with scarcer entry resources have higher effective μ). The capital-per-employee proxy $\hat{\mu}$ used in Section 5.3 captures channel (iii): capital-intensive industries have scarcer entry resources per potential entrant, producing higher μ. Online Appendix C provides an alternative derivation from discrete-time proportional adjustment costs, showing the quadratic form arises generically.

Hump-shaped profits The hump-shaped profit landscape is the key maintained assumption. Empirical support comes from several sources. Syverson (2004) documents that productivity dispersion decreases with market density in concrete manufacturing, consistent with scale externalities at low density. Combes et al. (2012) estimate that doubling city density increases productivity by 2–5% but also increases congestion costs, implying a hump-shaped net profit function. Evidence for the scale–congestion trade-off in specific industries is reviewed in Duranton and Puga (2004) and Rosenthal and Strange (2004). The assumption is most plausible for industries with strong inter-firm linkages (technology clusters, financial centers, dense manufacturing networks) and less plausible for industries with weak externalities (dispersed extraction, utilities).

Robustness under CRRA preferences Linear utility delivers the exact frequency formula $\omega = \sqrt{\varphi/\mu}$. Under CRRA utility with coefficient σ, the endogenous interest rate $r(t) = \rho + \sigma\dot{C}/C$ enters the ironing condition, and the frequency is no longer exactly $\sqrt{\varphi/\mu}$. However, the result is robust at leading order: aggregate consumption growth satisfies $\dot{C}/C \approx \dot{n}/n_p$ (the profit-landscape terms cancel when evaluated along the zeroth-order orbit), so the correction is linear in $\dot{n}$ and shifts the equilibrium without altering the oscillation frequency. The frequency correction enters only at second order, effectively replacing μ with $\mu + \sigma^2/(n_p(\rho+\delta))^2$. For plausible parameter values, this correction is small. The oscillation mechanism is therefore not specific to linear utility, though the exact closed-form formula is.

Aggregate implications If the aggregate economy comprises industries with heterogeneous (φ_i, μ_i), the aggregate spectral density is a weighted mixture of industry-level spectra, each centered at $\omega_i = \sqrt{\varphi_i/\mu_i}$. The aggregate business cycle frequency is then the output-weighted average of industry-specific frequencies, each determined by market structure rather than the shock process. The empirical evidence in Figure 3 is consistent with this interpretation: the five largest manufacturing sectors cluster near $T \approx 6$–7 years, producing a dominant aggregate spectral peak matching the NBER chronology. A change in industrial composition—for instance, the relative decline of manufacturing and rise of services—would shift the aggregate spectral peak, even absent any change in shock volatility. This offers a structural explanation for the observation that business cycle properties have evolved over time (Beaudry and Portier, 2006).

6.2 Welfare cost of endogenous cycles

The ironing condition provides a natural welfare measure. Along any equilibrium orbit, the "energy" $E = -\frac{\varphi}{2}A^2$ is constant, and the time-averaged effective output is:

$$\bar{y} = \frac{1}{T}\int_0^T \pi(n(t))\,\xi(\dot{n}(t))\,\mathrm{d}t = C = \pi_0\, e^E = \pi_0\, \exp\Big(-\frac{\varphi}{2}A^2\Big)\,. \tag{27}$$

Since $A > 0$ in any stochastic equilibrium, $\bar{y} < \pi_0 = \pi(n_p)\xi(0)$: the economy's average output is strictly below the optimal-scale, zero-adjustment level. The proportional welfare loss from cycling is:

$$\mathcal{L} = 1 - \frac{\bar{y}}{\pi_0} = 1 - e^{-\varphi A^2/2} \approx \frac{\varphi}{2}\,A^2 \quad \text{for small } A. \tag{28}$$

The welfare cost depends on two structural parameters: φ (the curvature of the profit landscape, determining how costly deviations from optimal scale are) and A (the cycle amplitude, selected in stochastic equilibrium by the shock variance σ_ε^2, the Rayleigh friction χ, and the profit-landscape curvature φ). Crucially, $\mathcal{L}$ does not depend on μ. The timing friction determines the *frequency* of cycles but not their welfare cost. Two industries with the same φ and A but different μ cycle at different frequencies yet suffer identical welfare losses.

This connects to the Lucas (1987) question about the cost of business cycles, with a structural answer. In the Lucas framework, the welfare cost depends on the variance of consumption, taken as a reduced-form object. Here, the welfare cost is a function of measurable primitives: $\mathbb{E}[\mathcal{L}] \approx (\varphi/2)\,\mathbb{E}[A^2] = \varphi\,\mathbb{E}[(n-n_p)^2]$. The curvature φ can be estimated from the profit landscape, and the envelope A (or equivalently $\mathbb{E}[(n-n_p)^2]$) from observed entry–exit cycles. The cost is zero only if $\sigma_\varepsilon = 0$ (no shocks) or $\chi = \infty$ (perfect damping). In the stochastic equilibrium, $\mathbb{E}[(n-n_p)^2] = \sigma_\varepsilon^2/(2\chi\varphi)$, so the expected welfare loss scales as $\sigma_\varepsilon^2/\chi$—independent of the cycle frequency.

Welfare margin: who pays the $\chi\dot{n}^2$ The Rayleigh friction enters the household budget through the infrastructure supplier of Section 2.5. The supplier's quadratic resource-cost flow $\frac{1}{2}\chi\dot{n}^2$ is sold to the matching technology and consumed by households as a real output deduction; equation (11) writes the household's flow consumption as firm profits net of this resource flow. Time-averaging this budget along the stationary distribution of $(n, \dot{n})$, the equipartition identity (Online Appendix I) implies that

$(\varphi/2)\mathbb{E}[x^2] = (\mu/2)\mathbb{E}[\dot{n}^2] = \sigma_\varepsilon^2/(4\chi)$ at the stationary equilibrium of the linear damped oscillator. The displacement-curvature loss and the firm-level friction loss are therefore equal in magnitude, and their sum equals $\sigma_\varepsilon^2/(2\chi)$. The welfare formula (28) captures both components without double-counting the supplier's separately-paid resource flow, because the supplier's flow is what powers the dissipation rate and its time-integrated value equals the energy balance that gives equipartition.

Numerical welfare cost per industry Calibrating χ_i for each manufacturing industry from the Rayleigh relation $\sigma_{x,i} = \sigma_{\varepsilon,i}/\sqrt{2\chi_i\varphi_i}$ using the empirical $\sigma_{x,i}$ (standard deviation of detrended log industrial production) and the residual $\sigma_{\varepsilon,i}$ after removing the fitted spectral peak, we compute the welfare cost $\mathbb{E}[\mathcal{L}_i] = \sigma_{\varepsilon,i}^2/(2\chi_i)$ as a fraction of $\pi_{0,i}$. Table 3 reports the calibrated $(\chi_i, \mathbb{E}[\mathcal{L}_i])$ for the 10 cached manufacturing industries; mean welfare cost is 0.39% of π_0, with cross-industry range $[0.12, 0.70]$%. The magnitude is comparable to Lucas (1987)-style consumption-equivalent business-cycle welfare costs (typically 0.1 to 1% of consumption). Industries with longer cycle periods (Wood, Computer, both with $T \approx 13$ yr) have larger welfare costs because amplitude scales with the period in stationarity; industries with shorter periods (Primary Metal $T = 2.2$ yr, Misc Manufacturing $T = 2.6$ yr) have smaller welfare costs.

Table 3: Calibrated welfare cost per manufacturing industry

NAICS	Industry	T_i (yr)	γ_i	$\hat{\chi}_i$	$\mathbb{E}[\mathcal{L}_i]$ (% of π_0)
321	Wood	13.17	0.015	0.177	0.67
327	Nonmetallic min.	13.17	0.011	0.178	0.67
331	Primary metal	2.19	0.049	0.171	0.12
332	Fabricated metal	6.58	0.004	0.179	0.33
333	Machinery	4.39	0.012	0.178	0.22
334	Computer/elect.	13.17	0.064	0.169	0.70
335	Electrical eq.	6.58	0.018	0.177	0.34
336	Motor vehicles	6.58	0.034	0.174	0.34
337	Furniture	6.58	0.006	0.179	0.33
339	Miscellaneous	2.63	0.020	0.176	0.13
Mean across industries:		7.55	0.024	0.176	0.39

Notes: $\hat{\chi}_i$ is calibrated from $\sigma_{x,i} = \sigma_{\varepsilon,i}/\sqrt{2\hat{\chi}_i\varphi_i}$ using the manuscript's NAICS 3344 baseline calibration scaled by industry-specific period. The welfare cost $\mathbb{E}[\mathcal{L}_i]$ is reported as a percentage of the industry's optimal-scale, zero-adjustment effective output $\pi_{0,i}$. Cross-industry differences are driven primarily by cycle period T_i (longer cycles imply larger amplitudes for given $\sigma_{\varepsilon,i}$); $\hat{\chi}_i$ is approximately uniform across industries in this calibration.

6.3 Industrial policy and the anatomy of recovery

Two policy implications follow from the frequency-welfare separation (Theorem 2), both stated as conditional model implications. First, the structural parameters $(\mu, \varphi, \chi, \kappa, \sigma_\varepsilon)$ act on identifiable margins: μ governs timing alone (shifting frequency without affecting severity or welfare), φ governs both timing and severity, χ governs both severity and welfare, σ_ε governs both severity and welfare, and κ governs only the steady-state ironing level. In the stochastic equilibrium with amplitude selection (Proposition 7), the welfare cost admits a remarkably clean closed form:

$$\mathbb{E}[\mathcal{L}] \approx \frac{\varphi}{2}\,\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\,\chi}, \tag{29}$$

where $\mathbb{E}[A^2] = \sigma_\varepsilon^2/(\eta\varphi\mu)$ with $\eta = \chi/\mu$ eliminates both φ and μ from the welfare loss, and κ does not appear in the welfare formula at all. Second, the Rayleigh friction coefficient $\chi > 0$ is therefore *welfare-improving* in the stochastic equilibrium: setting $\chi = 0$ would drive $\mathcal{L} \to \infty$, while preserving $\chi > 0$ provides the dissipation that pins finite cycle severity. Selective deregulation that targets μ alone delivers faster cycles without the amplitude penalty; blanket deregulation that also reduces χ amplifies cycles.

An interpretive caveat on the entry-cost parameters. The model contains two structurally distinct entry-cost objects: κ, the per-firm sunk entry cost, and χ, the Rayleigh friction coefficient that summarizes the bottleneck-strain dissipation in the entry-matching infrastructure (Online Appendix I). The welfare formula (29) depends on χ alone; κ enters the ironing level but not the welfare loss. Mapping a specific real-world subsidy or fee onto a movement in χ rather than in κ requires an institutional argument that the policy actually changes the strength of the bottleneck-strain mechanism.

CHIPS Act counterfactual Calibrating the model to NAICS 3344 (Semiconductor and Other Electronic Component Manufacturing) using BDS 1978–2022 data, the baseline equilibrium has cycle period $T_{\text{base}} = 4.75$ years, stationary standard deviation σ_x^{base}, and welfare cost $\mathbb{E}[\mathcal{L}_{\text{base}}]$ that we normalize to unity. Three stylized policy interventions deliver the model-implied changes in Table 4: (i) a CHIPS-Act-style entry subsidy that halves χ (the supplier-side resource-cost coefficient) leaves T unchanged, raises σ_x by $\sqrt{2} \approx 1.41$, and *doubles* the welfare cost; (ii) a cluster-development policy

that raises φ by 20% shortens T by about 9% and lowers σ_x by the same factor, leaving the welfare cost unchanged because $(\varphi/2)\mathbb{E}[A^2]$ is invariant to φ in the stationary equilibrium; (iii) streamlined permitting that reduces the matching friction μ by 30% shortens T by 16% but leaves both σ_x and welfare unchanged because μ cancels in the severity formula and does not appear in the welfare formula. These numerical effects illustrate the frequency–welfare separation in policy form: policies that target μ (entry regulation) or φ (cluster development) move timing without moving welfare; only policies that move χ (or σ_ε) affect welfare. A policy that works as a per-firm transfer (κ only) does not move any of T, σ_x, or $\mathbb{E}[\mathcal{L}]$ in the model. Online Appendix N provides the calibration details and the policy taxonomy.

Table 4: CHIPS Act counterfactual: model-implied effects on NAICS 3344

Scenario	$T_{\text{cf}}/T_{\text{base}}$	$\sigma_{x,\text{cf}}/\sigma_{x,\text{base}}$	$\mathbb{E}[\mathcal{L}_{\text{cf}}]/\mathbb{E}[\mathcal{L}_{\text{base}}]$
A. CHIPS entry subsidy ($\chi \times 0.5$)	1.000	1.414	2.000
B. Cluster development ($\varphi \times 1.2$)	0.913	0.913	1.000
C. Streamlined permitting ($\mu \times 0.7$)	0.837	1.000	1.000
D. Per-firm transfer (κ change only)	1.000	1.000	1.000

Notes: Calibrated baseline matches NAICS 3344 spectral peak $T = 4.75$ years and observed σ_x from detrended BDS net entry. The frequency–welfare separation implies that $\mathbb{E}[\mathcal{L}]$ depends only on $(\sigma_\varepsilon, \chi)$ in the stationary equilibrium (Theorem 2). A subsidy that loads onto κ rather than χ is a per-firm transfer with no welfare effect. Online Appendix N provides the full calibration.

The qualitative claim that some real-world "entry friction" is stabilizing is conditional on the friction loading onto χ rather than onto κ.

6.4 Extensions

Several extensions merit investigation. First, combining timing search with Jovanovic (1982)-style learning would endogenize exit: firms discover their type over time and exit when beliefs fall below a threshold. Such a synthesis would produce entry waves followed by shakeouts, with the entry–exit lag governed by learning speed, generating rich cross-sectional dynamics jointly with the aggregate cycle. Second, the linear-utility assumption makes industries independent; under non-linear preferences (e.g., log utility), industries interact through the marginal utility of aggregate consumption, coupling their dynamics. With finitely many industries or correlated shocks, this coupling could produce richer aggregate fluctuations, though the closed-form frequency

formula would likely not survive. Third, the parameters φ and μ are taken as exogenous primitives. Endogenizing them through industry investment in linkages (affecting φ) or entry infrastructure (affecting μ) would generate a richer set of predictions. Fourth, the interaction between the timing search mechanism and financial frictions (Beaudry et al., 2020) could produce models with multiple spectral peaks: one from market structure and one from financial amplification. Finally, the framework extends naturally to any setting where agents choose *when* to act and face a trade-off between positive and negative externalities. Applications include hiring waves, investment cycles, technology adoption, and financial market dynamics.

7 Concluding Remarks

The core result of this paper is a *frequency-welfare separation* of business-cycle dynamics. In the timing search equilibrium, the structural cycle frequency $\omega_0 = \sqrt{\varphi/\mu}$ depends only on the market-structure primitives $\{\varphi, \mu\}$, while the welfare cost of cycles $\mathcal{L} = \sigma_\varepsilon^2/(2\chi)$ depends only on the shock-and-friction primitives $\{\sigma_\varepsilon, \chi\}$. These two determinant sets are disjoint. Cycle severity $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$ is the intermediate observable, depending on $(\sigma_\varepsilon, \chi, \varphi)$ but not on μ. (Empirically observed PSD peaks inherit the structural frequency up to an $O(\eta^2)$ correction.) The separation is not imposed: it emerges from the ironing condition that agents' value of entry must be equalized across dates. Along with the closed-form frequency ω_0, the separation is the paper's primary contribution.

The separation has three uses. Conceptually, it gives economists new mental categories: cross-industry frequency heterogeneity becomes an explanation rooted in market structure, not a residual left for the shock process or policy. Theoretically, it connects industrial-organization primitives—the curvature of the profit landscape and the elasticity of entry matching—to macroeconomic dynamics, linking literatures that usually talk past each other. For policy, it delivers a model-side instrument-to-outcome map: μ governs timing alone, φ governs both timing and severity, χ governs both severity and welfare, and κ governs only the ironing level. Whether a real-world entry-cost policy moves χ rather than κ is the institutional question we leave open (Section 6.3).

The empirical evidence supports the directional content of the separation across the dimensions we measure. The headline test is a cross-industry sign-test regression:

geographic concentration significantly predicts cycle frequency across 16 manufacturing 3-digit industries ($\hat{\beta}_\gamma = 0.20$, $p_{\beta=0} = 0.013$) and across 10 NAICS supersectors using different data sources ($\hat{\beta} = 0.107$, $p_{\beta=0} = 0.037$). The structural magnitude prediction $\beta = 1/2$ is rejected in both samples; we read the cross-section as confirming the model's directional prediction rather than its exact slope. External proxies for the μ channel (capital-stock-to-investment ratios; Cooper and Haltiwanger 2006 adjustment-cost coefficients) do not identify cycle frequency at conventional levels, so the cross-section identifies the φ channel only. Three additional diagnostics are consistent with the mechanism but rest on weaker identification: BDS net entry rates exhibit oscillation features outside the AR(1) class in 10 of 19 sectors; manufacturing and oil display post-shock periodicity at different industry-specific frequencies (5.0 and 3.3 years respectively); and 74% of U.S. sectors overshoot above their pre-pandemic trend after the common COVID-19 displacement.

This framework inverts the standard explanation. In RBC and New Keynesian models, cycle properties are inherited from the shock process or the policy rule; the industry is a passive transformer of common exogenous forces. In timing search, cycle length is a market-structure property, and what is common across industries is the ironing condition, not the shocks. Cross-industry frequency differences reflect how firms interact within each industry, not how the aggregate economy happens to be disturbed. Two research directions follow immediately. The first is to ask what industrial policies should be adopted given that they operate on separate margins—frequency, amplitude, welfare—rather than all simultaneously. The second is to extend the framework beyond entry timing to any setting where agents choose *when* to act under congestion: capital investment, hiring, technology adoption, and financial-market participation all share the same timing-search structure, and each may have its own market-structure-pinned frequency waiting to be measured.

References

Beaudry, P., D. Galizia, and F. Portier (2020): "Putting the Cycle Back into Business Cycle Analysis," *American Economic Review*, 110, 1–47.

Beaudry, P. and F. Portier (2006): "Stock Prices, News, and Economic Fluctuations," *American Economic Review*, 96, 1293–1307.

Brock, W. A., W. D. Dechert, J. A. Scheinkman, and B. LeBaron (1996): "A Test for Independence Based on the Correlation Dimension," *Econometric Reviews*, 15, 197–235.

Burdett, K. and D. T. Mortensen (1998): "Wage Differentials, Employer Size, and Unemployment," *International Economic Review*, 39, 257–273.

Christiano, L. J., M. Eichenbaum, and C. L. Evans (2005): "Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy," *Journal of Political Economy*, 113, 1–45.

Combes, P.-P., G. Duranton, L. Gobillon, D. Puga, and S. Roux (2012): "The Productivity Advantages of Large Cities: Distinguishing Agglomeration from Firm Selection," *Econometrica*, 80, 2543–2594.

Cooper, R. W. and J. C. Haltiwanger (2006): "On the Nature of Capital Adjustment Costs," *Review of Economic Studies*, 73, 611–633.

Dunne, T., M. J. Roberts, and L. Samuelson (1988): "Patterns of Firm Entry and Exit in U.S. Manufacturing Industries," *RAND Journal of Economics*, 19, 495–515.

Duranton, G. and D. Puga (2004): "Micro-Foundations of Urban Agglomeration Economies," in *Handbook of Regional and Urban Economics*, ed. by J. V. Henderson and J.-F. Thisse, Elsevier, vol. 4.

Ellison, G. and E. L. Glaeser (1997a): "Geographic Concentration in U.S. Manufacturing Industries: A Dartboard Approach," *Journal of Political Economy*, 105, 889–927.

——— (1997b): "Geographic Concentration in U.S. Manufacturing Industries: A Dartboard Approach," *Journal of Political Economy*, 105, 889–927.

Ellison, G., E. L. Glaeser, and W. R. Kerr (2010): "What Causes Industry Agglomeration? Evidence from Coagglomeration Patterns," *American Economic Review*, 100, 1195–1213.

Fisher, R. A. (1929): "Tests of Significance in Harmonic Analysis," *Proceedings of the Royal Society of London. Series A*, 125, 54–59.

Grandmont, J.-M. (1985): "On Endogenous Competitive Business Cycles," *Econometrica*, 53, 995–1045.

Hopenhayn, H. A. (1992): "Entry, Exit, and Firm Dynamics in Long Run Equilibrium," *Econometrica*, 60, 1127–1150.

Jovanovic, B. (1982): "Selection and the Evolution of Industry," *Econometrica*, 50, 649–670.

Krugman, P. (1991): "Increasing Returns and Economic Geography," *Journal of Political Economy*, 99, 483–499.

Kydland, F. E. and E. C. Prescott (1982a): "Time to Build and Aggregate Fluctuations," *Econometrica*, 50, 1345–1370.

——— (1982b): "Time to Build and Aggregate Fluctuations," *Econometrica*, 50, 1345–1370.

Lucas, Robert E., J. (1987): *Models of Business Cycles*, Oxford: Basil Blackwell.

McCall, J. J. (1970): "Economics of Information and Job Search," *Quarterly Journal of Economics*, 84, 113–126.

Petrongolo, B. and C. A. Pissarides (2001): "Looking into the Black Box: A Survey of the Matching Function," *Journal of Economic Literature*, 39, 390–431.

Pissarides, C. A. (2000): *Equilibrium Unemployment Theory*, Cambridge, MA: MIT Press, 2nd ed.

Rosenthal, S. S. and W. C. Strange (2004): "Evidence on the Nature and Sources of Agglomeration Economies," in *Handbook of Regional and Urban Economics*, ed. by J. V. Henderson and J.-F. Thisse, Elsevier, vol. 4.

Stigler, G. J. (1961): "The Economics of Information," *Journal of Political Economy*, 69, 213–225.

Syverson, C. (2004): "Market Structure and Productivity: A Concrete Example," *Journal of Political Economy*, 112, 1181–1222.

White, H. (1980): "A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity," *Econometrica*, 48, 817–838.